

72nd BPSC PRELIMS

प्रहार

Test Series

Test-18/2

FULL LENGTH TEST

Solution

English



BPSC TEST-18/2 (SOLUTION)

Q 1. Ans. D

Explanation:

- Captain William Hawkins arrived at Jahangir's court in 1608, making him the first official English representative to visit a Mughal emperor. He stayed until around 1611 but ultimately failed to secure a trade treaty, largely due to Portuguese opposition and Jahangir's lukewarm interest.

Q 2. Ans. C

Explanation:

- Nabin Chandra Sen, the 19th-century Bengali poet, famously described the aftermath of the Battle of Plassey (1757) as a night of eternal darkness or gloom for India — capturing the sense that Plassey marked the definitive turning point after which Indian sovereignty was extinguished and colonial subjugation began.
- This is a fairly specific literary-historical fact that appears in standard Modern History sources (including Bipan Chandra's works) as an illustration of how later Indian intellectuals retrospectively understood the significance of Plassey.

Q 3. Ans. B

Explanation:

- When Warren Hastings ended the Dual Government (Clive's system of 1765) in 1772, the core change was that the Company stopped outsourcing administration through the Nawab's machinery and instead took direct control. Under the Dual system, the Company held the diwani (revenue rights) but left nizamat (law and order, administration) nominally with the Nawab — creating a situation where the Company had power without responsibility. This was widely blamed for the catastrophic Bengal Famine of 1770.
- Ending it meant Company servants directly supervised revenue collection and administration, replacing the Indian intermediary apparatus.

Q 4. Ans. D

Explanation:

- Most historians regard the Battle of Plassey (1757) as marking the beginning of the Drain of Wealth from Bengal. After Siraj-ud-Daulah's defeat, the East India Company and its officials extracted enormous

sums through gifts, plunder, and political influence over the Nawab. The process intensified after the Grant of Diwani rights in 1765, when the Company obtained the right to collect revenue from Bengal, Bihar, and Orissa.

Q 5. Ans. A

Explanation:

- **Statement 1 is correct:** Lord Wellesley established Fort William College in Calcutta in 1800 to train Company recruits in Indian languages, law, and administration after their arrival in India. It was indeed the initial training institution for civil servants entering Bengal.
- **Statement 2 is correct:** The Court of Directors in London was uncomfortable with Wellesley's grand vision for Fort William College — it was expensive and outside their direct control. They curtailed its role and established Haileybury College in England in 1806 as the primary training institution, where recruits would be trained before coming to India. Haileybury thus effectively replaced Fort William College as the central training institution, though Fort William continued in a reduced capacity for language training.
- **Statement 3 is incorrect:** This is the trap. Competitive examinations did not govern Civil Service recruitment throughout the Company rule. For most of the Company period, recruitment was based on patronage and nomination — the Directors of the Company had the privilege of nominating candidates. It was only after the Charter Act of 1853 that competitive examinations were introduced for Civil Service recruitment, and this came right at the very end of Company rule (which ended in 1858).

Q 6. Ans. B

Explanation:

- **Bahadur Shah I (1707–1712):** Aurangzeb's son, won the war of succession after his father's death in 1707. Had a relatively stable reign, made peace with the Rajputs and Marathas.
- **Jahandar Shah (1712–1713):** Came to power after Bahadur Shah I's death through a succession struggle, with the backing of the powerful noble Zulfiqar Khan. His reign lasted barely a year.
- **Farrukh Siyar (1713–1719):** Overthrew Jahandar

Shah with the crucial support of the Sayyid Brothers (Abdullah Khan and Husain Ali Khan). His reign is notable for the 1717 farman granting trading privileges to the English East India Company. He was eventually deposed and killed by the Sayyid Brothers.

- **Muhammad Shah (1719–1748):** After two short-lived puppet emperors installed by the Sayyid Brothers, Muhammad Shah came to the throne, eventually freed himself from Sayyid dominance, and had a long reign — though it included the catastrophic Nadir Shah invasion of 1739.

Q 7. Ans. B

Explanation:

- As a young man, he was sent to Varanasi (Benares), the preeminent center of Hindu scholarship, where he deeply studied Sanskrit grammar, the Vedas, the Upanishads, and classical Hindu philosophical systems.
- Patna was a renowned center of Islamic learning during the late 18th century. Rammohun Roy studied Persian and Arabic there, engaging thoroughly with the Quran, Islamic theology, Sufi mystic poetry, and Arabic translations of Aristotle and Euclid.
- While Rammohun Roy did learn ancient Greek and Hebrew to read the Old and New Testaments in their original languages, he studied these languages in Calcutta, not Paris. He did visit France later in life (in 1832), but his major scriptural linguistic studies were conducted entirely in India.

Q 8. Ans. A

Explanation:

- **Refusal of greased cartridges (April 24, 1857):** Eighty-five skirmishers of the 3rd Native Cavalry at Meerut openly refused to touch or bite the newly introduced Enfield rifle cartridges, believing they were greased with cow and pig fat.
- **Imprisonment of eighty-five soldiers (May 9, 1857):** The defiant soldiers were court-martialed, stripped of their uniforms, shackled with irons in front of the entire garrison, and sentenced to 10 years of rigorous imprisonment. This public humiliation deeply angered their comrades.
- **Outbreak of the Revolt at Meerut (May 10, 1857):** The very next day (Sunday evening), the Indian sepoys at Meerut mutinied. They broke open the jail, released their imprisoned comrades, killed British officers, set fire to their bungalows, and raised the banner of revolt.

- **Death of Rani Jhansi (June 17, 1858):** Rani Lakshmibai of Jhansi fought a long, heroic campaign through late 1857 and 1858. She was ultimately killed in battle near the Kotah-ki-Serai area near Gwalior, over a year after the initial outbreak at Meerut.

Q 9. Ans. B

Explanation:

- **Statement 1 is correct:** King Alaungpaya (reigned 1752–1760) is celebrated as one of the three greatest monarchs in Burmese history. During the mid-eighteenth century, he successfully launched a rapid military campaign that overthrew the Mon rulers of the south, subdued various rival ethnic factions, and reunified Burma, establishing the Third Burmese Empire under the Konbaung Dynasty.
- **Statement 2 is incorrect:** Alaungpaya did not rule from Ava. He was originally a village headman from the town of Moksobomyo, which he renamed Shwebo (located northwest of Ava, between the Mu and Irrawaddy rivers). He established Shwebo as his royal capital and ruled from there throughout his entire reign. While his army successfully recaptured Ava from the Mon forces in 1753, the capital of the empire was moved back to Ava by his successors later on.
- **Statement 3 is correct:** Alaungpaya was the foundational patriarch and first king of the Konbaung Dynasty. King Bodawpaya (reigned 1782–1819) was the sixth king of the same dynasty (and one of Alaungpaya's sons). Therefore, Alaungpaya's reign directly and chronologically preceded that of Bodawpaya.

Q 10. Ans. C

Explanation:

- Surendranath Banerjea (often called the Rashtraguru) began his public career in Calcutta (Kolkata) during the mid-1870s after being unjustly dismissed from the Indian Civil Service (ICS) by the colonial government.
- Upon his return to India, he joined Vidyasagar College as a professor of English. From this academic platform, he began delivering a series of powerful, stirring nationalist lectures to student communities in Calcutta. He spoke passionately about Indian unity, the regular exploitation under British rule, and drew inspiration from European nationalist movements, introducing young students to the lives of leaders like Giuseppe Mazzini and the Italian unification.

Q 11. Ans. C

Explanation:

- Dr. Rajendra Prasad, son of Mahadev Sahai, was born on 3 December 1884 at Zeradei, Siwan (Bihar). Married to Rajvanshi Devi at the age of 12, he later became the first President of India and served from 1950 to 1962. He was awarded the Bharat Ratna in 1962. His notable works include Satyagraha at Champaran, India Divided, Atmakatha, Mahatma Gandhi and Bihar: Some Reminiscences, and Babu ke Kadmon Mein. He spent his final years at Sadaqat Ashram, Patna, and passed away on 28 February 1963.
- Maulana Abul Kalam Azad, a prominent freedom fighter, scholar, and independent India's first Education Minister (1947–1958), played a pivotal role in shaping the country's modern education system. He was instrumental in the establishment of the first Indian Institute of Technology (IIT) at Kharagpur in 1951 and the University Grants Commission (UGC) in 1953 to promote higher education and research. In recognition of his immense contribution to nation-building and education, he was posthumously awarded the Bharat Ratna in 1992. His birth anniversary, 11 November, is observed annually as National Education Day across India.
- Jayaprakash Narayan (1902–1979), popularly known as “Lok Nayak”, was a renowned freedom fighter, socialist leader, and advocate of democratic values. He led the historic Sampooran Kranti (Total Revolution) Movement in the 1970s against corruption and authoritarianism. He received the Ramon Magsaysay Award for Public Service in 1965 and was posthumously awarded the Bharat Ratna in 1999 for his outstanding contributions to the nation.
- Ustad Bismillah Khan (1916–2006) was a legendary shehnai maestro who elevated the instrument from a traditional ceremonial role to the stage of Indian classical music. Born in Dumraon, Bihar, he became closely associated with Varanasi (Kashi) and was deeply devoted to the Ganga and the city's cultural heritage. He famously performed the shehnai from the Red Fort on 15 August 1947, marking India's Independence. He received all four of India's highest civilian honours—Padma Shri (1961), Padma Bhushan (1968), Padma Vibhushan (1980), and Bharat Ratna (2001).

Q 12. Ans. A

Explanation:

- Statement 1 is correct: Longitudinal extension remains the same. After the creation of Jharkhand on 15 November 2000, Bihar lost its southern districts. The easternmost and westernmost boundaries remained virtually unchanged. Present Bihar extends from 83°19' 50' E to 88°17' 40' E longitude. Since Jharkhand was carved out mainly from the southern portion of undivided Bihar, the east-west spread (longitude) remained the same.
- Statement 2 is incorrect: Latitudinal extension reduced by 2°30'. Bihar extends from 24°20' 10' N to 27°31' 15' N latitude. Before bifurcation, undivided Bihar extended approximately from 21°58' 10' N to 27°31' 15' N. Actual reduction ≈ 2°22', not 2°30'.
- Statement 3 is correct: It comprises 54.15% area of undivided Bihar. Presently, Bihar has an area of 94,163 sq km. Area of undivided Bihar before bifurcation: 173,877 sq km. Thus, present Bihar retained 54.15% of the area of undivided Bihar.

Q 13. Ans. A

Explanation:

- Patharkatti Stone Craft, a traditional stone-carving art practiced for nearly 300 years in Patharkatti village of Gaya district, Bihar, was granted a Geographical Indication (GI) tag in 2025. Renowned for its intricate sculptures crafted from distinctive blue-black granite, the craft sustains the livelihoods of more than 650 artisans. The GI recognition helps preserve the authenticity of the craft and safeguards its rich artistic heritage.
- Sujni (Sujani) Embroidery, a traditional embroidery craft of Bihar, received the Geographical Indication (GI) tag in 2006. Practiced primarily in Bhusra village of Muzaffarpur district and selected villages of Madhubani district, this unique handicraft is known for its intricate needlework and storytelling motifs. The GI registration under the Geographical Indications of Goods (Registration and Protection) Act, 1999 provides legal protection to the craft and its artisans.
- Bhagalpuri Silk, one of Bihar's most celebrated textile traditions, was awarded the Geographical Indication (GI) tag in 2013 under the Handicraft/Textile category. Produced in Bhagalpur district, often referred to as the “Silk City of India,” this silk is renowned for its quality and craftsmanship. The

GI tag protects the intellectual property rights of local weavers and ensures that only silk produced within the designated geographical region can be marketed as **Bhagalpuri Silk**.

Q 14. Ans. B

Explanation:

- “**Abua Raj Seter Jana, Maharani Raj Tundu Jana**” (also expressed as “**Abua Dishum, Abua Raj**”) was the powerful slogan of **Bhagwan Birsa Munda (1875–1900)**, the iconic tribal freedom fighter, social reformer, and leader of the **Munda Ulgulan (Great Rebellion)** against British rule in the Chotanagpur region. In the Mundari language, the slogan means “**Let our own rule prevail; let the Queen’s rule end,**” reflecting the tribal aspiration for self-governance and freedom from colonial oppression. Revered as “**Dharti Aba**” (**Father of the Earth**), Birsa Munda inspired resistance against exploitation and the alienation of tribal lands. Although he died in British custody on **9 June 1900** at the age of 25, his struggle left a lasting legacy, culminating in the enactment of the **Chotanagpur Tenancy Act, 1908**, which safeguarded tribal land rights by restricting the transfer of tribal land to non-tribals.

Q 15. Ans. D

Explanation:

- The first major incident of the Revolt of 1857 in Bihar occurred on **12 June 1857** at Rohini, near Deoghar (present-day Jharkhand, then part of Bihar). This event is regarded as the earliest spark of the uprising in the Bihar region. The rebellion took place at the military headquarters located at Rohini, where soldiers stationed under British command turned against their officers. During the incident, British officers **Lieutenant Sir Norman Leslie** and **Assistant Surgeon Dr. Grant** were attacked; Leslie was killed while Grant was seriously injured. The revolt was subsequently suppressed by **Major MacDonald**, the commanding officer.
- The significance of the Rohini incident lies in the fact that it marked the **beginning of the Revolt of 1857 in Bihar**, preceding the more famous uprisings at Patna under **Pir Ali Khan (3 July 1857)** and **Danapur (25 July 1857)**, which later joined forces with the movement led by **Veer Kunwar Singh**.

Q 16. Ans. D

Explanation:

- The **Champaran Agrarian Enquiry Committee**

was constituted by the Government of Bihar and Orissa in **June 1917** to investigate the exploitation of indigo cultivators under the **Tinkathia system** in Champaran. The seven-member committee was chaired by **F.G. Sly** and included **Mahatma Gandhi, L.G. Adami, G. Rainy, E.L. Tanner, Raja Harihar Prashad Narayan Singh, and D.J. Reid**. Gandhi represented the grievances of the peasants (ryots), while the committee also included government officials, landlords, and European planters. Its recommendations led to the **abolition of the Tinkathia system** and the enactment of the **Champaran Agrarian Act, 1918**, marking a major success of Gandhi’s first Satyagraha in India.

Q 17. Ans. B

Explanation:

- The **Bihar Provincial Kisan Sabha (BPKS)** was established in **1929** under the leadership of **Swami Sahajanand Saraswati** to mobilize peasants against zamindari exploitation and agrarian injustices. Conceived at the **Sonepur Mela** in Bihar, the organization emerged as the leading platform for peasant activism in the province and was supported by prominent leaders such as **Jadunandan Sharma and Karyanand Sharma**. The widespread success of the BPKS in organizing rural masses laid the foundation for the creation of the **All India Kisan Sabha (AIKS)** at the **Lucknow Session of the Indian National Congress in 1936**, with **Swami Sahajanand Saraswati** playing a central role in its formation and serving as its first president.

Q 18. Ans. C

Explanation:

- **Bihar State Credit & Investment Corporation Ltd. (BICICO):** Established in **1975**, the **Bihar State Credit & Investment Corporation Ltd. (BICICO)** is a state government financial institution functioning under the **Department of Industries, Government of Bihar**. It provides financial assistance and support to **small, medium, and large-scale industrial units** in the state to promote industrial development. Its headquarters is located on **Bailey Road, Patna**.
- The **Bihar Industrial Area Development Authority (BIADA)** is a statutory body constituted under the **Bihar Industrial Area Development Authority Act, 1974**. Functioning under the **Department of Industries, Government of Bihar**, it is responsible for promoting planned industrial growth by developing and managing industrial infrastructure

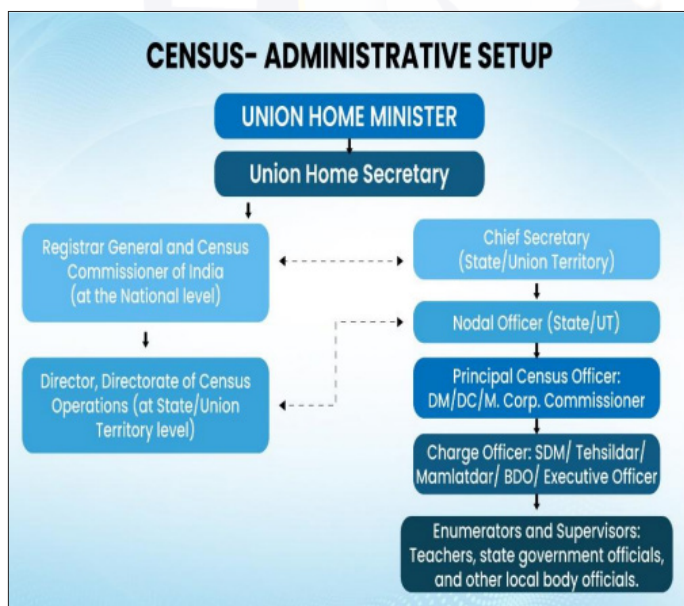
across the state. BIADA oversees **more than 90 industrial areas, industrial estates, and industrial parks**, which are organized into **North and South Zones** with major industrial clusters at **Patna, Bihta, Muzaffarpur, Gaya, and Bhagalpur**.

- **Bihar State Tourism Development Corporation (BSTDC):** Established in 1980, BSTDC functions under the **Tourism Department, Government of Bihar** and is responsible for tourism promotion, management of tourist facilities, and development of tourism infrastructure across the state
- The **Bihar State Housing Board** is a statutory authority constituted under the **Bihar State Housing Board Act, 1982**. It is responsible for developing housing schemes and related infrastructure to provide affordable housing across the state.

Q 19. Ans. A

Explanation:

- **Statement 1 is incorrect:** The earliest references to population enumeration in India are found in **Kautilya's Arthashastra (321–296 BCE)**. Detailed records of population and administrative statistics were later documented by **Abul Fazl** in the **Ain-i-Akbari** during the reign of **Akbar**.



- **Statement 2 is incorrect:** **Census 2027** will be the **16th Census** conducted in the history of India since the first non-synchronous census operations began in 1872 and the first synchronous census was conducted in 1881. It will also be the **8th Census after Independence**, as seven censuses have already been conducted in independent India (1951, 1961, 1971, 1981, 1991, 2001 and 2011).
- **Statement 3 is correct:** According to the

Constitution, Census is a Union subject. Therefore, the Census is conducted and coordinated by the **Government of India** through the Office of the Registrar General and Census Commissioner. However, its field operations are carried out with the assistance of **State Governments and Union Territory administrations**, ensuring uniform implementation across the country.

Q 20. Ans. A

Explanation:

- **Assertion (A) is true** The occurrence of Piedmont Swamp Soil in Bihar is linked to the Himalayan foothill environment. Piedmont Swamp Soils are found in the **northwestern and northeastern fringes of Bihar**, particularly in districts adjoining the **Shiwalik (Sub-Himalayan) foothills**, such as West Champaran and parts of the Kosi-Mahananda region. These soils occur in the **Tarai and foothill belt**, where poor drainage and seasonal waterlogging create swampy conditions.
- **Reason (R) is true:** Piedmont zones develop where sediments accumulate at the base of mountain fronts. A **piedmont** is a gently sloping zone formed at the foot of mountains. Rivers descending from the Himalayas lose velocity upon entering the plains and deposit large quantities of **sand, silt, gravel, and other sediments**. This depositional environment creates the piedmont belt.
- **Why is (R) the correct explanation of (A)?**
 - ✓ The Himalayan rivers transport huge sediment loads from the mountains. As these sediments accumulate at the mountain front, they form **piedmont plains**. In Bihar's Tarai region, the high groundwater table, poor drainage, and frequent flooding in these depositional zones lead to the development of **Piedmont Swamp Soils**.

Q 21. Ans. C

Explanation:

- **Statement 1 is correct:** The National Health Authority (NHA) and the Government of West Bengal signed a formal Memorandum of Understanding (MoU) in New Delhi. With this signing, West Bengal officially became the 36th State/Union Territory to adopt the program, effectively completing its pan-India coverage across all administrative territories.
- **Statement 2 is correct:** The fundamental structural offering of AB-PMJAY is providing a cashless and paperless health insurance cover of ₹5 lakh per family

per year. This coverage is specifically designed to handle catastrophic health expenditures involving secondary and tertiary care hospitalization through a vast network of empanelled public and private healthcare facilities.

Q 22. Ans. B

Explanation:

- **Statement 1 is correct:** The Gyan Bharatam Mission was launched under the Ministry of Culture and aims to survey, document, and preserve over one crore ancient Indian manuscripts nationwide. Statement 1 accurately describes it as a nationwide manuscript heritage survey.
- **Statement 2 is incorrect:** The survey is not a three-month exercise. The Gyan Bharatam Mission was launched in the Union Budget 2025-26 with a total budget of more than ₹490 crore for 2025–2031 — it is a six-year long-term mission, not a three-month survey. Statement 2 is factually wrong. ThePrint
- **Statement 3 is incorrect:** The Gyan Bharatam Mission has nothing to do with repatriation of antiquities. Its focus is on locating manuscripts through a national survey, documenting every manuscript repository for a National Electronic Database, conserving manuscripts, and training scholars in manuscript studies. Repatriation of antiquities is an entirely separate policy matter.

Q 23. Ans. B

Explanation:

- Cabinet-approved with a total budget allocation of ₹10,371.92 crore, the IndiaAI Mission is implemented by the IndiaAI Independent Business Division (IBD) under the Digital India Corporation (MeitY). The mission is structurally anchored across seven distinct core pillars:
 - ✓ **IndiaAI Datasets Platform (AIKosh):** (Component 1) A centralized, unified national platform that streamlines access to high-quality, non-personal datasets, open models, and toolkits for Indian startups, researchers, and public institutions.
 - ✓ **IndiaAI Compute Capacity:** An ambitious public-private partnership (PPP) model establishing a scalable AI computing ecosystem with a target cluster of 10,000 or more high-end Graphics Processing Units (GPUs) to offer affordable AI-as-a-Service on a public cloud.
 - ✓ **Safe and Trusted AI:** This framework develops indigenous tools, self-assessment compliance

checklists, and evaluation frameworks to minimize bias and combat deepfakes. It includes the establishment of the dedicated IndiaAI Safety Institute (AIS) to enforce responsible and ethical AI deployment.

- ✓ **IndiaAI Innovation Centre:** Focuses on the development and deployment of indigenous Large Multimodal Models (LMMs) and sector-specific foundational pipelines.
- ✓ **IndiaAI Application Development Initiative:** Targets high-impact social sectors (such as healthcare, agriculture, governance, and climate resilience) by taking challenge statements directly from public departments and engineering targeted solutions.
- ✓ **IndiaAI FutureSkills:** Aimed at expanding AI education pathways by augmenting PhD fellowships, postgraduate programs, and setting up localized Data & AI Labs across Tier-2 and Tier-3 cities. IndiaAI Startup
- ✓ **Financing:** Streamlines risk-capital access for deep-tech AI startups to help accelerate them from concepts into marketable products.
- **The National Quantum Grid** is part of the completely separate National Quantum Mission (NQM) (under the Department of Science and Technology), which focuses on developing quantum computing, communication, and cryptography systems. It is not an administrative pillar of the IndiaAI Mission.

Q 24. Ans. B

Explanation:

- Fujairah Port is located in the Emirate of Fujairah, United Arab Emirates (UAE) on the country's eastern coast along the Gulf of Oman.
 - ✓ It is the only major UAE port situated outside the Persian Gulf and the Strait of Hormuz, giving it immense strategic significance.
 - ✓ The port serves as a key bunkering, oil storage, and transshipment hub, allowing maritime traffic to access the Indian Ocean without passing through the Strait of Hormuz.
 - ✓ Its location enhances the UAE's energy security and global trade connectivity.

Q 25. Ans. C

Explanation:

- **Statement 1 is correct:** The mission is structured with a hybrid funding approach. The core technology platforms, national registries, and

cloud infrastructure (like AgriStack and Krishi-DSS) are funded 100% by the Centre as Central Sector components. However, the ground-level execution—such as conducting the Digital Crop Survey (DCS) and state-specific IT integrations—is funded under a Centrally Sponsored model, with a typical 60:40 split between the Centre and general states (90:10 for North Eastern and Himalayan states).

- **Statement 2 is incorrect:** The mission does not mandate direct procurement. Instead of acting as a procurement agency, the mission creates the enabling infrastructure (verified farmer IDs, land maps, and crop data). While this authenticated data can be utilized by procurement portals to streamline their operations, the mission itself contains no statutory mandate forcing digital procurement.
- **Statement 3 is correct:** One of the primary objectives of building this DPI is to foster an ecosystem where the government, private sector, and startups can build farmer-centric solutions. By creating a unified registry, it enables states to facilitate trusted digital access for farmers to services like localized weather alerts, credit, insurance, and the secure online purchase and sale of agricultural inputs and produce.

Q 26. Ans. D

Explanation:

- The 43rd session of the UNESCO General Conference, the governing body of the United Nations Educational, Scientific and Cultural Organization (UNESCO), commenced in Samarkand, Uzbekistan. This marked the first time in four decades that the General Conference was held outside UNESCO's headquarters in Paris.
- The General Conference, which meets biennially, brings together representatives of UNESCO's Member States to determine the organisation's policies, approve its programme and budget, and deliberate on major issues related to education, culture, science, and communication.

Q 27. Ans. C

Explanation:

- **Statement 1 is correct:** The Sujal Gaon ID was launched by the Union Minister of Jal Shakti under the flagship Jal Jeevan Mission (JJM). It specifically aligns with the transition toward "Sujalam Bharat"—focusing on service delivery, real-time monitoring, and sustainable water management rather than just infrastructure creation.

- **Statement 2 is correct:** The Sujal Gaon ID acts as a scheme-based unique digital identifier for every rural piped drinking water supply scheme in the country. It essentially creates a complete digital footprint by mapping the entire rural water supply system from its source down to the household tap.

Q 28. Ans. B

Explanation:

- NITI Aayog's Frontier Tech Hub, in collaboration with IBM Corporation as its designated knowledge partner, recently unveiled a comprehensive national roadmap titled "Transforming India into a Leading Quantum-Powered Economy."

Q 29. Ans. D

Explanation:

- **Statement 1 is correct:** The Annual Survey of Unincorporated Sector Enterprises (ASUSE) is conducted by the National Statistical Office (NSO) under the Ministry of Statistics and Programme Implementation (MoSPI).
- **Statement 2 is correct:** The survey covers the unincorporated non-agricultural sector, including enterprises engaged in manufacturing, trade, and services.
- **Statement 3 is correct:** ASUSE provides crucial inputs for GDP estimation and the compilation of National Accounts, besides generating information on indicators such as employment, Gross Value Added (GVA), wages, and digital usage.

Q 30. Ans. B

Explanation:

- The Meerut Bugle, which recently received a Geographical Indication (GI) tag, has long been associated with India's military tradition. Meerut's bugle-making heritage dates back to the late nineteenth century, when these instruments were used as command tools during the British era.
- Over time, the bugle retained its importance in regimental bands, military drills, ceremonial parades, and national events. Even today, Meerut-made bugles are widely used by the Indian Army, paramilitary forces, police units, and several military and police academies.

Q 31. Ans. D

Explanation:

- The Unique Identification Authority of India (UIDAI) launched the 'Aadhaar Vision 2032'

framework as a comprehensive 10-year strategic and technological roadmap. All four advanced technologies are integrated as core structural pillars of this upgrade:

- ✓ **Artificial Intelligence (AI):** It focuses on facilitating smart identity verification, faster processing for concurrent users, robust anomaly detection, and automated fraud prevention.
- ✓ **Blockchain technology:** The framework introduces blockchain integrations to enhance trust, transparency, and immutability in specific digital transaction trails, establishing a highly secure and tamper-proof log ecosystem.
- ✓ **Quantum Computing:** To protect the national identity database against upcoming cyber threats powered by advanced computing breakthroughs, the framework initiates a transition toward quantum-resilient security systems and quantum-safe cryptography.
- ✓ **Advanced Encryption Techniques:** This operates on a "Privacy-by-Design" approach that directly complies with the principles of the Digital Personal Data Protection (DPDP) Act by prioritizing consent-bound data usage and minimal data retention.

Q 32. Ans. C

Explanation:

- The Fission Surface Power (FSP) Project is a joint initiative led by NASA and the U.S. Department of Energy (DOE) to design, build, and deploy a compact, automated nuclear fission reactor on the lunar surface. It forms a key technological foundation for the Artemis program's goal of establishing a long-term, sustainable presence on the Moon.

Q 33. Ans. B

Explanation:

- **Justice Revati Mohite Dere (Meghalaya High Court):** Formerly a senior judge of the Bombay High Court, she was elevated and sworn in as the Chief Justice of the Meghalaya High Court. She succeeded Justice Soumen Sen upon his transfer.
- **Justice Sangam Kumar Sahu (Patna High Court):** Previously a senior judge serving at the Orissa High Court, he was elevated to head the judiciary in Bihar, taking his oath of office as the 47th Chief Justice of the Patna High Court.

- **Justice Mahesh Sharadchandra Sonak (Jharkhand High Court):** Formerly a judge of the Bombay High Court (notable for registering Goa's first 'Living Will'), he was appointed and sworn in as the 18th Chief Justice of the Jharkhand High Court, succeeding Justice Tarlok Singh Chauhan.
- **Justice Soumen Sen (Kerala High Court):** Originally a senior judge from the Calcutta High Court who briefly served as the Chief Justice of the Meghalaya High Court, he was subsequently transferred and sworn in as the 40th Chief Justice of the Kerala High Court.

Q 34. Ans. B

Explanation:

- **Udaipur Jheel → Bihar**
 - ✓ Udaipur Jheel is a freshwater wetland located in Udaipur district of Bihar. It is an important habitat for migratory waterbirds and supports rich aquatic biodiversity. The common misconception is to associate it with Udaipur city in Rajasthan, but this is an entirely different and lesser-known site in Bihar.
- **Khichan Wetland → Rajasthan**
 - ✓ Khichan Wetland is located near Phalodi town in Jodhpur district of Rajasthan. It is world-famous for hosting tens of thousands of Demoiselle Cranes (*Grus virgo*), locally called Kurja, during the winter migration season. The local Marwari community has traditionally fed and protected these cranes for centuries, making it a unique example of community-driven wildlife conservation.
- **Chhari Dhand → Gujarat**
 - ✓ Chhari Dhand is a seasonal wetland situated within the Rann of Kutch in Gujarat. It serves as a critical habitat for flamingos, pelicans, and numerous other migratory waterbirds. It falls within the Kutch Desert Wildlife Sanctuary and is ecologically significant as one of the few permanent water bodies in an otherwise arid landscape.
- **Patna Bird Sanctuary → Uttar Pradesh**
 - ✓ Patna Bird Sanctuary is located in Etah district of Uttar Pradesh and must not be confused with the city of Patna in Bihar. It is centred around a shallow seasonal lake and serves as an important wintering ground for migratory ducks, geese, and wading birds arriving from Central Asia and Siberia.

Q 35. Ans. A**Explanation:**

- **Statement 1 is correct:** The 20th G20 Leaders' Summit, scheduled to be held in 2025 in South Africa, marks the first-ever G20 summit hosted on the African continent.
- **Statement 2 is correct:** The overarching theme adopted for the South African G20 presidency is "Solidarity, Equality and Sustainability", focusing heavily on addressing the developmental priorities of the Global South, poverty, inequality, and systemic global economic reforms.
- **Statement 3 is incorrect:** The summit is hosted under the national presidency of South Africa, not the African Union (AU). While the African Union participates fully as a permanent member (admitted during the New Delhi Summit in 2023), the annual rotating presidency belongs to the host country, South Africa.

Q 36. Ans. A**Explanation:**

- **Statement 1 is correct:** The Belan Valley (Uttar Pradesh) is an important archaeological site where excavations show a continuous cultural sequence from Lower, Middle to Upper Palaeolithic phases, making it significant for understanding stratified prehistoric occupation in India.
- **Statement 2 is incorrect:** Lower Palaeolithic tools such as hand-axes and cleavers were generally made from coarse-grained quartzite and other locally available hard stones, not specifically fine-grained cryptocrystalline quartzite. The emphasis was on durability and availability rather than fine-grained material.

Q 37. Ans. A**Explanation:**

- **Statement 1 is correct:** The Great Bath at Mohenjodaro is a marvel of ancient engineering. To make the brick tank completely watertight, the Harappans laid the bricks on edge using a high-quality gypsum mortar. To prevent any subsoil water seepage, they applied a thick, foundational layer of natural bitumen (tar) along the sides and floor of the pool.
- **Statement 2 is incorrect:** The statement completely reverses the mathematical logic of the Harappan weight system. For lower fractional denominations, the weights followed a binary system (doubling each time: \$1, 2, 4, 8, 16, 32, 64\$, all the way up

to \$12,800\$). The weight of \$16\$ acted as the traditional baseline unit (roughly equivalent to \$13.64\$ grams). For higher, larger denominations, the system shifted into a decimal system (such as weights of \$160, 200, 320, 640, 1600, 3200, 6400\$, etc.).

Q 38. Ans. A**Explanation:**

- In the Mauryan administrative framework detailed in Kautilya's Arthashastra, the state exercised tight control over the economy through a network of specialised department superintendents known as Adhyakshas.
 - ✓ **Sitadhyaksha:** Managed and supervised state-owned crown agricultural lands, known as Sita. This officer was responsible for regulating cultivation, organising labour (including slaves and prisoners), and maximising agricultural yield.

Q 39. Ans. C**Explanation:**

- **Statement 1 is correct:** In the early Rig Vedic period, bali was a purely voluntary offering or tribute brought by tribal members to their chief (Rajan). However, as society transitioned into the Later Vedic phase, settled agriculture took root, and large territorial kingdoms (Janapadas) emerged. To support the growing royal machinery and administrative needs, this voluntary gift became institutionalised into a mandatory, regular tax collected by a specialised officer called the Bhagadugha.
- **Statement 2 is correct:** The Satapatha Brahmana is a massive, highly detailed prose text from the Later Vedic era. It contains explicit references to agrarian expansion, specifically describing continuous rituals of ploughing with heavy, ox-drawn iron ploughshares to break up the dense, sticky alluvial soil of the upper and middle Gangetic plain.

Q 40. Ans. B**Explanation:**

- **Statement 1 is correct:** The early Rig Vedic society was overwhelmingly pastoral and nomadic. Agriculture played a secondary role. Livestock, specifically the cow (Gau), was the central measure of wealth, the medium of exchange, and the primary cause of tribal conflicts (which were fittingly called Gavishti—literally meaning "a search for cows").

- **Statement 2 is incorrect:** The nishka was not a state-minted coin or currency. During the early Rig Vedic age, there was no centralised state structure or administrative machinery capable of minting currency. The nishka was a prestigious gold ornament or necklace used by elite members of society. While it did sometimes serve as a convenient medium of exchange or unit of value in transactions, it was a prestige item rather than official metallic money.

Q 41. Ans. D

Explanation:

- **Maha-dandanayaka → Chief Judicial Officer**
 - ✓ The Maha-dandanayaka was a high-ranking judicial authority responsible for maintaining law and order and delivering justice across the empire.
- **Sandhivigrahika → Minister of War and Peace**
 - ✓ The Sandhivigrahika handled foreign affairs, including diplomacy, alliances, treaties, and declarations of war. This office was crucial in imperial administration.
- **Uparika → Provincial Governor**
 - ✓ The Uparika served as the provincial governor appointed directly by the king to administer large territorial divisions known as Bhuktis.
- **Kumaramatyā → High-ranking Administrative Officer**
 - ✓ The Kumaramatyā belonged to an elite cadre of senior civil servants from which top ministers, governors, and military commanders were often selected.

Q 42. Ans. B

Explanation:

- **Assertion (A) is true:** The post-Gupta period (from roughly the 6th century CE onward) did experience a steep decline in direct maritime trade with the Western Roman Empire. In fact, by this time, the Western Roman Empire itself had politically collapsed.
- **Reason (R) is true:** As detailed in R.S. Sharma's thesis on early Indian feudalism, the decentralisation of state power and the massive proliferation of royal land grants led to the rise of localised, autonomous, self-sufficient village economies. This subcontinental shift did naturally diminish the vibrant domestic demand and network for external luxury goods.
- **Why (R) is NOT the correct explanation for (A):**

- ✓ The dramatic contraction of long-distance commerce with the Western Roman Empire was driven by external geopolitical factors, not by India's internal village structure.
- ✓ The primary cause was the external breakdown of the Roman economy, along with the subsequent expansion of the Byzantine Empire (Eastern Roman Empire) which learned the secret of silk cultivation directly from China around the 6th century CE.

Q 43. Ans. D

Explanation:

- **Pair 1 is incorrectly matched: Sohan Valley ' Upper Palaeolithic**
 - ✓ It is one of the prominent sites of the Lower Palaeolithic age in the Indian subcontinent (located in the Siwalik region of present-day Pakistan). It is famous for the 'Soanian culture', which is characterized by primitive pebble-based chopper and chopping stone tools.
- **Pair 2 is incorrectly matched: Gufkral ' Mesolithic**
 - ✓ Located in the Kashmir Valley, Gufkral (literally translating to 'cave of the potter') is a major Neolithic site. Much like the nearby site of Burzahom, it is known for its distinctive pit dwellings, bone tools, and early evidence of agriculture and animal husbandry.
- **Pair 3 is incorrectly matched: Patne ' Neolithic**
 - ✓ Located in the Tapi river basin in Maharashtra, Patne is primarily renowned for its Upper Palaeolithic (and subsequent Mesolithic) horizons. It is particularly famous in Indian archaeology for the discovery of ostrich eggshell pieces, some of which feature engraved cross-hatched designs, providing rare evidence of Upper Palaeolithic art and bead-making in India.

Q 44. Ans. A

- **At the time of Aurangzeb's death in 1707, the Mughal Empire reached its greatest territorial extent, but behind this massive facade, the administrative and military machineries were on the verge of collapse:**
 - ✓ **Administrative Decay:** Because Aurangzeb spent the final 25 years of his life personally commanding military campaigns in the Deccan (the "Deccan Ulcer"), he completely neglected the administration of northern India. The central authority weakened, provincial

governors began acting semi-independently, and the Jagirdari crisis (an acute shortage of lands to pay nobles) severely fractured the loyalty of the bureaucracy.

- ✓ **Military Breakdown:** The endless war with the Marathas emptied the imperial treasury. The army was physically exhausted, heavily overstretched, and demoralized. Because of severe financial bankruptcy, soldiers' salaries were frequently delayed for months or even years, leading to frequent mutinous tendencies and indiscipline among the troops.

Q 45. Ans. A

Explanation:

- **Bahadur Shah I → Complete exhaustion of the 13 crore royal treasury accumulated by Aurangzeb**
 - ✓ During the reign of Bahadur Shah I (1707–1712), prolonged succession struggles and military campaigns led to the depletion of the vast treasury inherited from Aurangzeb.
- **Jahandar Shah → Wholesale introduction of the oppressive Ijarah (revenue farming) system**
 - ✓ Under Jahandar Shah (1712–1713), the Ijarah system gained prominence, allowing revenue collection rights to be farmed out to contractors, often resulting in exploitation of the peasantry.
- **Muhammad Shah → Catastrophic invasion and looting by Nadir Shah**
 - ✓ During the reign of Muhammad Shah (1719–1748), Nadir Shah of Persia invaded India in 1739, sacked Delhi, and carried away immense wealth, including the Peacock Throne and the Koh-i-Noor.
- **Shah Alam II → Formal loss of Delhi to the British Army**
 - ✓ During the reign of Shah Alam II (1759–1806), the British captured Delhi in 1803 during the Second Anglo-Maratha War, bringing the Mughal emperor under British protection.

Q 46. Ans. D

Explanation:

- **Statement 1 is correct:** The Sakoli Series of the Dharwar System is explicitly mapped as stretching over the Jabalpur and Rewa districts. Because both of these administrative districts lie entirely within the geographical boundaries of Madhya Pradesh, the series is described as being confined to this state.
- **Statement 2 is correct:** The Sakoli Series is noted for containing abundant mica-bearing rocks and is

regarded as one of the mica-rich Dharwarian series in India.

- **Statement 3 is correct:** Lithologically, the Sakoli Series is highly metamorphic. Its rock composition is predominantly a mix of mica-schists, quartzites, dolomite, and marble. The marble sourced from this particular series in the Jabalpur region is historically prized for its excellent quality.

Q 47. Ans. C

Explanation:

- **Shivalik Fossil Park (Himachal Pradesh):** Also known as the Suketi Fossil Park, it is located in the Sirmaur district of Himachal Pradesh. It is famous for its collection of prehistoric vertebrate fossils and skeletons restored from the upper Shivalik sedimentary strata.
- **Mandla Plant Fossils National Park (Madhya Pradesh):** Located in the Mandla district of Madhya Pradesh, this national park preserves excellent plant fossils that existed in India between 40 million and 150 million years ago, primarily embedded during the formation of the Deccan Traps.
- **Akal Wood Fossil Park (Rajasthan):** Located in the Jaisalmer district of Rajasthan, this park features petrified wood logs and fossils dating back to the Lower Jurassic period (around 180 million years ago), standing as stark evidence of the lush, humid forests that once stood in what is now the Thar Desert.
- **National Fossil Wood Park (Tamil Nadu):** Located at Tiruvakkarai (Villupuram district) and Sathanur (Perambalur district) in Tamil Nadu, these geo-heritage sites preserve massive petrified tree trunks that date back to the Cretaceous period.

Q 48. Ans. A

Explanation:

- **Pair A is incorrect:** Khardung La is located on the Ladakh Range, north of Leh, and serves as the gateway to the Shyok and Nubra valleys. It is absolutely not the same as Qara-Tagh-La. Qara-Tagh-La (Black Pass) is an entirely separate, historical high-altitude pass located much further north on the actual Karakoram Range, along the ancient silk route connecting Ladakh with Yarkand.

Other Mountain Passes

- **Banihal Pass:** Situated on the Pir Panjal range, it connects the Jammu region with the Kashmir Valley. The famous Jawahar Tunnel was constructed beneath this pass to ensure all-weather road connectivity.

- **Zoji La:** Located on the Great Himalayas range, this vital pass serves as the ethno-geographical gateway connecting Srinagar (Kashmir Valley) with Kargil and Leh (Ladakh).
- **Pensi La:** Known as the “Gateway to Zaskar,” this high mountain pass connects the Suru Valley (Kargil district) with the Zaskar Valley in Ladakh, historically considered part of the broader Kashmir Himalayan system.

Q 49. Ans. B

Explanation:

- **Assertion (A) is correct:** The Himalayas do not terminate abruptly. At their western and eastern ends, they exhibit sharp structural bends known as syntaxial bends. The western syntaxial bend near Nanga Parbat represents a dramatic turn of the Himalayan ranges around a pivotal point.
- **Reason (R) is also correct:** The Indus River is an antecedent river that predates the uplift of the Himalayas. As the Nanga Parbat massif rose, the river maintained its original course through rapid downcutting, carving the exceptionally deep Indus Gorge.
- **However, (R) is not the correct explanation of (A):** The formation of the Indus Gorge did not cause the Himalayan bend. The western syntaxial bend is tectonic in origin, resulting from the collision of the Indian and Eurasian Plates. The northwestern corner of the Indian Plate acted as a rigid pivot, around which the compressed Tethyan strata were folded, producing the characteristic hairpin-shaped bend of the Himalayas.

Q 50. Ans. B

Explanation:

- **Statement 1 is correct:** A barbed drainage pattern (also called a hooked drainage pattern) is a rare, anomalous network where tributaries run in a direction opposite to that of the master river. Consequently, when looking at a map, their confluences form sharp, acute angles that look like the barbs of a fishhook, making the tributaries appear to point upstream against the current flow.
- **Statement 2 is correct:** This layout is a classic geomorphological footprint of river capture (stream piracy) or tectonic tilting. When a more aggressive, lower-level stream breaches a water divide and captures the upper reaches of another river, the master river's flow is completely reversed. However, its existing tributaries are stubborn; they

retain their old geological orientation and continue flowing in their original direction, now running counter to the newly reversed main stream.

- **Statement 3 is correct:** In the Indian subcontinent, the Arun River of Nepal (an antecedent river that forms part of the Saptakoshi system) is example of a barbed drainage pattern. Complex structural uplifts and river captures within the Himalayan folds forced course reversals that left its upper tributaries joining at highly discordant, barbed angles.
- **Statement 4 is incorrect:** Barbed drainage is not a subtype of trellis drainage. Trellis drainage develops in regions of alternating bands of hard and soft rocks (folded strata) where subsequent streams meet the main stream at right angles. Barbed drainage, on the other hand, is an independent, discordant pattern defined fundamentally by its unique directional clash (upstream-pointing acute junctions) resulting from river capture or rapid tectonic tilting, rather than differential erosion along parallel fault lines.

Q 51. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Chenab River is known as Asikni (or Asikni/Askani) in Sanskrit literature. It is the Ravi River that is historically known as Parushini (or Iravati). It is called Chandra-Bhaga in Himachal Pradesh because it is formed by the confluence of two mountain streams, the Chandra and the Bhaga, at Tandi near Keylong.
- **Statement 2 is correct:** After originating in Himachal Pradesh, the Chenab enters Jammu and Kashmir (specifically the Kishtwar district) as a roaring mountain torrent. It cuts a magnificent, deep gorge directly through the rugged Pir Panjal range near Kishtwar before flowing past Doda and Ramban, ultimately entering the plains of Punjab in Pakistan near Sialkot.

Q 52. Ans. B

Explanation:

- **Peat** represents the precursor stage of coal formation and consists of partially decayed plant debris. It is not regarded as true coal and contains more than 75% moisture with a carbon content generally below 40–50%.
- **Lignite (Brown Coal)** is the lowest rank of actual coal. It is soft, often retains a distinct woody texture, and possesses a carbon content of about 60–70% along with high moisture content.
- **Bituminous Coal** is the most common and widely

used variety of coal. It is dense and black, usually displaying alternating shiny and dull bands, and contains approximately 75–85% carbon, giving it a high calorific value.

- **Anthracite (Hard Coal)** is the highest rank of coal. It is jet-black, metallic in appearance, and exceptionally hard, with a carbon content exceeding 85–95% and the lowest moisture and volatile matter among coal types.

Q 53. Ans. D

Explanation:

- **Statement 1 is correct:** Most extra-peninsular (Himalayan) rivers — the Indus, Ganga, Brahmaputra and their tributaries — originate from Himalayan glaciers and snowfields, making them perennial. This is a defining characteristic distinguishing them from peninsular rivers which are largely rain-fed.
- **Statement 2 is correct:** In the plains, Himalayan rivers flow over gentle gradients with large volumes of water, making them broadly navigable. The Ganga system in particular has historically been one of India's most important navigable waterways.
- **Statement 3 is correct:** In terms of the fluvial cycle of erosion, extra-peninsular rivers are geologically young and active. Because the Himalayas are still tectonically uplifting, these streams are dynamically cutting downwards, forming steep V-shaped valleys, rapids, and deep gorges in their upper reaches. This places them squarely in the youthful stage (specifically the late youth stage as they transition from high-velocity mountain cutting to mature valley widening).
- **Statement 4 is correct:** Extra-peninsular (Himalayan) rivers carry heavy silt loads, have gentle gradients near the coast, and deposit enough sediment to build deltas (Ganga-Brahmaputra delta being the world's largest).

Q 54. Ans. B

Explanation:

- **Cumulonimbus clouds are characterised by extremely powerful updrafts and downdrafts. Large hailstone formation depends precisely on this vertical dynamics:**
 - ✓ A water droplet is carried high into the freezing zone of the cloud by strong updrafts, where it freezes.
 - ✓ It then partially falls back into warmer, supercooled water droplet zones, gaining a new layer of ice.

- ✓ The updraft carries it back up again, adding another ice layer.
- ✓ This repeated cycling above and below the freezing level builds up concentric layers of ice which is why cutting a large hailstone in cross-section reveals onion-like rings, each representing one cycle.
- ✓ Eventually the hailstone becomes too heavy for the updraft to support and falls to the ground.

Q 55. Ans. C

Explanation:

- **Statement 1 is correct:** The Government of India Act of 1935 abolished the dyarchy introduced in the provinces by the Act of 1919 and introduced provincial autonomy, under which provinces became autonomous units administered by responsible ministers answerable to provincial legislatures.
- **Statement 2 is correct:** The Act provided for the establishment of an All India Federation consisting of the British provinces and the princely states, though the federation never came into being because the requisite number of princely states did not accede.
 - ✓ The Act also introduced dyarchy at the Centre, which likewise never came into operation, abolished the Council of India established under the 1858 Act, extended franchise, provided for a Federal Court that was set up in 1937, and provided for the establishment of the Reserve Bank of India. It divided powers between the Centre and units through three lists, the Federal, Provincial, and Concurrent Lists, an arrangement that the Constitution later adopted in the Seventh Schedule.

Q 56. Ans. D

Explanation:

- **Statement 1 is correct:** Citizenship by birth is acquired by persons born in India within the periods and conditions specified in the Act, with the conditions tightened progressively, especially after the cut-off of 3 December 2004, when at least one parent must be a citizen and the other not an illegal migrant.
- **Statement 2 is correct:** Citizenship by descent is acquired by a person born outside India whose parent was a citizen of India at the time of birth, subject to registration at an Indian consulate within the prescribed period.

- **Statement 3 is correct:** Citizenship by naturalisation is granted to a qualified foreigner, who is not an illegal migrant, on application after fulfilling residence and other conditions and on taking an oath of allegiance.
- **Statement 4 is correct:** Citizenship by incorporation of territory arises when a new territory becomes part of India and the Government of India specifies the persons of that territory who shall be citizens. The fifth mode is acquisition by registration. The Act provides for loss of citizenship by renunciation, termination on voluntary acquisition of foreign citizenship, and deprivation by the government, and India follows single citizenship along with the scheme of Overseas Citizens of India, who are not full citizens and lack political rights.
- **Statement 2 is incorrect:** The Constitution does not use the expression judicial review anywhere, but the power is implied from several provisions such as Articles 13, 32, 131 to 136, 143, 226, 246, 251, 254, and 372.
- Judicial review is the power of the courts to examine the constitutionality of legislative enactments and executive orders of both the Centre and the states, and to declare them void if they contravene the Constitution. The Ninth Schedule was added by the 1st Amendment of 1951, and Article 31B granted immunity from judicial review to the laws placed in it. The scope of judicial review in India is narrower than in the United States, since Indian courts apply the test of procedure established by law rather than the broader American due process of law, though the latter has been read into Article 21 in substance.

Q 57. Ans. A

Explanation:

- **Article 40** directs the State to organise **village panchayats** and endow them with powers to function as units of self-government, and it is classified among the Gandhian principles. It provided the inspiration for the constitutional status later given to panchayats by the 73rd Amendment.
- **Article 44** directs the State to secure a **uniform civil code** for the citizens throughout the territory of India, and it is classified among the liberal intellectual principles.
- **Article 48A** directs the State to protect and improve the **environment and to safeguard forests** and wildlife, and it was one of the new directives added by the 42nd Amendment Act of 1976.
- **Article 50** directs the State to **separate the judiciary from the executive** in the public services of the State. The directive principles are non-justiciable and are classified into socialist, Gandhian, and liberal intellectual categories; other new principles added in 1976 were Articles 39(f), 39A on equal justice and free legal aid, and 43A on workers' participation in management, while the 97th Amendment added Article 43B on cooperative societies.

Q 58. Ans. A

Explanation:

- **Statement 1 is correct:** In the I. R. Coelho case of 2007 the Supreme Court held that laws placed in the Ninth Schedule after 24 April 1973 are open to judicial review if they violate fundamental rights forming part of the basic structure, so the protection of the Ninth Schedule is not absolute.

Q 59. Ans. C

Explanation:

- **Assertion is true:** The Election Commission has functioned as a multi member body consisting of the Chief Election Commissioner and two other Election Commissioners since October 1993, when the appointment of two Election Commissioners made it multi member on a continuing basis.
- **Reason is false:** The Constitution did not originally provide for a fixed multi member body; under Article 324 the Commission was to consist of the Chief Election Commissioner and such number of other Election Commissioners as the President may fix from time to time. The Commission therefore remained a single member body from 1950 until 1989, briefly became multi member in 1989, reverted to single member in 1990, and became multi member again from 1993.
- The Chief Election Commissioner and the other Election Commissioners enjoy equal powers and decide matters by majority. The Chief Election Commissioner can be removed only in the manner and on the grounds applicable to a judge of the Supreme Court, whereas the other Election Commissioners can be removed only on the recommendation of the Chief Election Commissioner, a distinction designed to protect the independence of the office.

Q 60. Ans. A

Explanation:

- **Statement 1 is correct:** Article 19 originally guaranteed seven freedoms, but the right to

acquire, hold, and dispose of property under Article 19(1)(f) was deleted by the 44th Amendment Act of 1978, leaving six freedoms, namely speech and expression, assembly, association, movement, residence, and profession.

- **Statement 2 is correct:** Each freedom is subject to reasonable restrictions that the State may impose on specified grounds, such as the sovereignty and integrity of India, public order, decency, morality, security of the State, and friendly relations with foreign states, and the reasonableness of a restriction is justiciable.
- **Statement 3 is incorrect:** The freedoms under Article 19 are available only to citizens and not to foreigners or to legal persons such as companies. The freedom of speech and expression has been judicially expanded to include the freedom of the press, the right to information, the right to silence, and the right to fly the national flag, while restrictions can be imposed on grounds including contempt of court, defamation, and incitement to an offence.

Q 61. Ans. A

Explanation:

- **Statement 1 is correct:** The Constitution of India is the lengthiest written constitution in the world, its bulk arising from the vast size and diversity of the country, a single constitution for both the Centre and the states, the dominance of legal experts in the Constituent Assembly, and the borrowing of detailed provisions from many sources including the Government of India Act of 1935.
- **Statement 2 is incorrect:** The amendment process is neither wholly rigid nor wholly flexible but a synthesis of both, since some provisions are amended by a simple majority, others by a special majority, and a few by a special majority along with ratification by half of the states.
- Other salient features include a blend of parliamentary sovereignty and judicial supremacy, an integrated and independent judiciary, a federal system with unitary or centralising features, fundamental rights and fundamental duties, directive principles, universal adult franchise, single citizenship, emergency provisions, three tier government after the 73rd and 74th Amendments, and the description of India as a sovereign socialist secular democratic republic. Critics have called it a borrowed constitution, a paradise of lawyers, and unduly bulky.

Q 62. Ans. A

Explanation:

- **Statement 1 is correct:** The Vice-President is elected by an electoral college consisting of the members of both Houses of Parliament, including the nominated members, through the system of proportional representation by means of a single transferable vote and by secret ballot. State legislatures play no part in this election, unlike in the election of the President.
- **Statement 2 is incorrect:** The Vice-President can be removed by a resolution of the Rajya Sabha passed by an effective majority and agreed to by the Lok Sabha, and no ground need be mentioned in the resolution, since the Constitution prescribes no grounds for removal.
- **Additional Information:** To be eligible a person must be a citizen of India, have completed thirty five years of age, and be qualified for election to the Rajya Sabha, and must not hold any office of profit. The Vice-President holds office for five years, acts as the ex officio Chairman of the Rajya Sabha, and discharges the functions of the President during a vacancy in that office for a maximum period of six months. The Indian Vice-President, unlike the American Vice-President, does not succeed to the office of President for the remainder of the term.

Q 63. Ans. B

Explanation:

- **Statement 1 is incorrect:** A constitution amendment bill under Article 368 can be introduced in either House of Parliament, not only in the Lok Sabha, and it can be introduced by a minister or a private member without the prior permission of the President.
- **Statement 2 is correct:** Amendments that affect the federal structure, such as the election of the President, the distribution of legislative powers, the representation of states in Parliament, or Article 368 itself, require ratification by the legislatures of half of the states by a simple majority, in addition to passage by Parliament.
- Each House must pass the bill by a special majority, that is a majority of the total membership of the House and a majority of two thirds of the members present and voting, and there is no provision for a joint sitting in case of disagreement. After passage and any required ratification, the President must give assent, since the 24th Amendment Act made

assent mandatory for constitution amendment bills. The Constitution can also be amended by a simple majority for certain matters outside the scope of Article 368, such as creation of new states and changes in citizenship.

Q 64. Ans. A

Explanation:

- **Statement 1 is correct:** The division of powers between the Centre and the states through the Union, State, and Concurrent Lists of the Seventh Schedule is a federal feature, since both levels of government derive their authority from the Constitution.
- **Statement 2 is incorrect:** Single citizenship is a unitary feature, not a federal one, because a true federation such as the United States provides for dual citizenship of the federation and the constituent states.
- **Statement 3 is correct:** An independent judiciary headed by the Supreme Court, with the power of judicial review to settle disputes between the Centre and the states and to interpret the Constitution, is a federal feature.
- **Statement 4 is incorrect:** The appointment of Governors by the Centre is a unitary feature, since the Governor acts as an agent of the Centre in the state. Other federal features are a written and rigid constitution and bicameralism, while other unitary features include a strong Centre, the absence of equality of state representation, the existence of All India Services, integrated machinery for elections and audit, and the emergency provisions; the framers therefore described the system as federal in form but unitary in spirit, and K. C. Wheare called it quasi federal.

Q 65. Ans. A

Explanation:

- **Golak Nath case → Held that Fundamental Rights cannot be amended**
 - ✓ In the Golak Nath case of 1967, the Supreme Court held that fundamental rights cannot be amended by Parliament, treating a constitution amendment as law within the meaning of Article 13 and therefore void if it abridged a fundamental right.
- **Kesavananda Bharati case → Propounded the Basic Structure Doctrine**
 - ✓ In the Kesavananda Bharati case of 1973, the Court overruled Golak Nath and held that Parliament can amend any part of the Constitution including fundamental rights,

but cannot alter its basic structure, thereby propounding the basic structure doctrine.

- **Minerva Mills case → Added judicial review to the Basic Structure**

- ✓ In the Minerva Mills case of 1980, the Court added the power of judicial review and the balance between fundamental rights and directive principles to the basic structure, and struck down clauses of the 42nd Amendment that sought to make amendments immune from review.

- **Waman Rao case → Fixed the cut-off date for the Basic Structure Doctrine**

- ✓ In the Waman Rao case of 1981, the Court clarified that the basic structure doctrine would apply to constitutional amendments enacted after 24 April 1973, the date of the Kesavananda Bharati judgment. The 24th Amendment had been enacted in response to Golak Nath to affirm Parliament's power to amend fundamental rights.

Q 66. Ans. A

Explanation:

- **Statement 1 is correct:** A prokaryotic cell, such as a bacterium or a cyanobacterium, lacks a true membrane-bound nucleus; its genetic material lies free in the cytoplasm in a region called the nucleoid.
- **Statement 2 is correct:** The cell wall of plant cells is made chiefly of cellulose, a complex carbohydrate that gives the cell rigidity and a definite shape.
- **Statement 3 is incorrect:** Animal cells have no cell wall and possess only small vacuoles or none. A large central vacuole is a feature of plant cells, not animal cells.
- **Additional facts:** Eukaryotic cells (plants, animals, fungi, protists) have a true nucleus and membrane-bound organelles. Robert Hooke first observed dead cell walls in cork, and the cell theory was given by Schleiden and Schwann.

Q 67. Ans. A

Explanation:

- **Assertion is true:** Metamaterials are engineered materials that can control light and other waves in ways ordinary materials cannot, including bending light the "wrong" way (negative refraction).
- **Reason is true:** These properties come from their carefully designed internal structure, made of repeating units smaller than the wavelength of the wave, rather than from the chemistry of the material.

- Since the unusual behaviour results directly from the engineered structure, the reason correctly explains the assertion.
- **Additional facts:** Metamaterials are explored for invisibility “cloaks,” superlenses that beat the normal resolution limit, compact antennas, and improved sensors.

Q 68. Ans. A

Explanation:

- **Thallophyta** → Simple plants lacking true roots, stems, and leaves
 - ✓ Thallophytes possess an undifferentiated plant body called a thallus. They do not have true roots, stems, or leaves. Examples include algae.
- **Bryophyta** → Mosses and liverworts
 - ✓ Bryophytes are non-vascular plants commonly known as the amphibians of the plant kingdom. Mosses and liverworts are their typical representatives.
- **Pteridophyta** → Plants with vascular tissue but no seeds
 - ✓ Pteridophytes are the first vascular land plants. They possess xylem and phloem but reproduce through spores rather than seeds. Examples include ferns and horsetails.
- **Phanerogams** → Plants that bear seeds
 - ✓ Phanerogams are seed-producing plants with well-developed reproductive structures and include both gymnosperms and angiosperms.

Q 69. Ans. B

Explanation:

- **Statement 1 is incorrect:** The structural and functional unit of the kidney is the nephron, not the neuron, which is the unit of the nervous system. Each human kidney has about a million nephrons.
- **Statement 2 is correct:** As the filtrate flows through the kidney tubule, useful substances such as glucose, amino acids, and most of the water and salts are reabsorbed into the blood.
- **Additional facts:** Urine formation involves filtration in the glomerulus, followed by selective reabsorption and secretion. Urine passes from the kidneys through the ureters to the urinary bladder and out through the urethra.

Q 70. Ans. A

Explanation:

- **Statement 1 is correct:** Rhizobium bacteria living in the root nodules of leguminous plants fix

atmospheric nitrogen into nitrogenous compounds that the plant can use.

- **Statement 2 is correct:** During lightning, the high temperature makes atmospheric nitrogen and oxygen combine into oxides of nitrogen, which dissolve in rain and reach the soil.
- **Statement 3 is incorrect:** Denitrifying bacteria convert soil nitrates back into free nitrogen gas, returning it to the atmosphere; they do not convert nitrates into ammonia.
- **Additional facts:** Nitrogen is essential for proteins and nucleic acids. Atmospheric nitrogen makes up about 78 per cent of air but is largely inert, becoming available mainly through biological fixation, lightning, and industrial fixation.

Q 71. Ans. A

Explanation:

- **Assertion (A) is true:** A Hypersonic Glide Vehicle (HGV) is generally harder to intercept than a traditional ballistic missile. Unlike ballistic missiles, which follow a largely predictable trajectory, HGVs can alter their flight path during the terminal phase.
- **Reason (R) is also true:** HGVs travel at hypersonic speeds (Mach 5 or higher) and are capable of manoeuvring while gliding through the atmosphere. This combination of extreme speed and unpredictable movement reduces the reaction time available to missile defence systems and complicates interception.
- **Additional facts:** Green plants (producers) form the first trophic level. The flow of energy is unidirectional, whereas nutrients are recycled. Harmful chemicals can become more concentrated up the chain, a process called biomagnification.

Q 72. Ans. A

Explanation:

- **Assertion is true:** a swimmer pushes the water backward with the hands and feet, which is the action force.
- **Reason is true:** by Newton's third law, the water pushes the swimmer forward with an equal and opposite reaction force, propelling the swimmer ahead. Since the forward motion is exactly the reaction to pushing the water backward, the reason correctly explains the assertion.
- **Additional facts:** The same law explains the recoil of a gun, the launch of a rocket (hot gases pushed backward), and walking, where the foot pushes the ground backward and the ground pushes the body forward.

Q 73. Ans. C**Explanation:**

- **Statement 1 is correct:** An object floats when its average density is less than that of the liquid, because then the buoyant force can balance its weight.
- **Statement 2 is correct:** A solid iron block sinks because iron is denser than water, but a ship is hollow, so its overall density is less than water and it displaces enough water to float.
- **Additional facts:** By Archimedes' principle, the upward buoyant force on a body equals the weight of the liquid it displaces. This principle is used in ships, submarines, and hydrometers.

Q 74. Ans. D**Explanation:**

- **Assertion is false:** temperature measures the degree of hotness, or the average kinetic energy of the particles, not the total heat content. A large tub of warm water can hold more heat than a tiny red-hot spark, despite its lower temperature.
- **The reason is true:** heat always flows from a body at higher temperature to one at lower temperature until both reach the same temperature (thermal equilibrium).
- **Additional facts:** Heat is a form of energy measured in joules, whereas temperature is measured in kelvin or degrees Celsius. The same amount of heat raises the temperature of different substances differently, depending on their specific heat capacity.

Q 75. Ans. C**Explanation:**

- **Statement 1 is correct:** For the same pair of surfaces, sliding (kinetic) friction is greater than rolling friction, which is why wheels and ball bearings are used to ease motion.
- **Statement 2 is incorrect:** Lubricants such as oil and grease reduce friction by forming a thin layer between the surfaces; they do not increase it.
- **Statement 3 is correct:** Friction is necessary; without it we could not walk, as our feet would slip, and brakes could not stop a moving vehicle.
- **Additional facts:** Friction also has drawbacks: it causes wear and tear and wastes energy as heat. It is increased by treads on tyres and shoe soles and reduced by polishing and lubrication.

Q 76. Ans. B**Explanation:**

- Dispersion is the splitting of white light into its

seven constituent colours (VIBGYOR) as it passes through a prism, because each colour is refracted, or bent, by a different amount.

- Refraction is the bending of light as it passes from one medium to another; dispersion occurs because of refraction but is not the splitting itself.
- Reflection is the bouncing back of light, and scattering is the spreading of light by tiny particles, which makes the sky appear blue.
- **Additional facts:** A rainbow forms by the dispersion of sunlight by water droplets. While passing through a prism, red light bends the least and violet light bends the most.

Q 77. Ans. C**Explanation:**

- **Statement 1 is correct:** During evaporation the more energetic particles escape, and the remaining liquid absorbs heat from its surroundings to make up for the lost energy, producing a cooling effect.
- **Statement 2 is incorrect:** The rate of evaporation decreases as humidity increases, because the surrounding air is already rich in water vapour and can hold little more.
- **Statement 3 is correct:** Gases diffuse much faster than liquids because their particles move rapidly, are far apart, and have very weak forces between them.
- **Additional facts:** Evaporation is a surface phenomenon that occurs at all temperatures, unlike boiling. The rate of evaporation increases with a rise in temperature, surface area, and wind speed.

Q 78. Ans. D**Explanation:**

- **Statement 1 is correct:** In a combination reaction, two or more reactants unite to form a single product, as when calcium oxide reacts with water to give calcium hydroxide.
- **Statement 2 is correct:** When a single compound breaks into two or more simpler substances on heating, the change is called thermal decomposition, as in the heating of calcium carbonate.
- **Statement 3 is incorrect:** In a redox reaction, oxidation and reduction occur simultaneously, not in separate steps; one substance loses electrons while the other gains them at the same time.
- **Statement 4 is correct:** Rancidity is the slow oxidation of fats and oils in food, which spoils their taste and smell; it is slowed by antioxidants and by airtight or nitrogen-flushed packaging.

- **Additional facts:** Reactions are also classified as displacement and double displacement. Corrosion, such as the rusting of iron, and rancidity are everyday examples of oxidation.

Q 79. Ans. B

Explanation:

- **Statement 1 is incorrect:** Unsaturated hydrocarbons contain at least one double bond (alkenes, such as ethene) or triple bond (alkynes, such as ethyne); having only single bonds is the feature of saturated hydrocarbons.
- **Statement 2 is correct:** Methane (CH_4) is the simplest saturated hydrocarbon (an alkane), containing only single covalent bonds between its atoms.
- **Additional facts:** Saturated hydrocarbons generally burn with a clean blue flame, whereas unsaturated hydrocarbons tend to burn with a yellow, sooty flame owing to incomplete combustion.

Q 80. Ans. C

Explanation:

- **Pair 1 is correct:** Burning magnesium in air combines magnesium and oxygen into a single product, magnesium oxide, which is a combination reaction.
- **Pair 2 is correct:** Heating calcium carbonate decomposes it into calcium oxide and carbon dioxide, a thermal decomposition reaction.
- **Pair 3 is correct:** Mixing barium chloride and sodium sulphate solutions forms an insoluble white precipitate of barium sulphate, a precipitation reaction.
- **Pair 4 is incorrect:** The digestion of food is the breakdown of large molecules into smaller ones, which is a decomposition, not a combination, process.
- **Additional facts:** A combination reaction releases energy when exothermic, such as quicklime reacting with water, while a decomposition reaction usually absorbs energy as heat, light, or electricity.

Q 81. Ans. D

Explanation:

- **Statement 1 is incorrect:** A candle flame is hottest in its outermost zone, the blue non-luminous region where combustion is complete; the innermost dark zone, containing unburnt wax vapour, is the least hot.
- **Statement 2 is incorrect:** The ignition temperature is the lowest temperature at which a substance catches fire, not the highest temperature at which it stops burning.

- **Additional facts:** A substance catches fire only when heated to its ignition temperature. Inflammable substances such as petrol have very low ignition temperatures. A fire is put out by removing fuel, heat, or oxygen.

Q 82. Ans. A

Explanation:

- **Assertion (A) is true:** A Hypersonic Glide Vehicle (HGV) is generally harder to intercept than a traditional ballistic missile. Unlike ballistic missiles, which follow a largely predictable trajectory, HGVs can alter their flight path during the terminal phase.
- **Reason (R) is also true:** HGVs travel at hypersonic speeds (Mach 5 or higher) and are capable of manoeuvring while gliding through the atmosphere. This combination of extreme speed and unpredictable movement reduces the reaction time available to missile defence systems and complicates interception.
- **Additional facts:** An HGV is boosted by a rocket and then glides unpowered to its target, unlike a cruise missile, which is powered throughout. Its low, weaving flight path also evades many radar and defence systems.

Q 83. Ans. D

Explanation:

- **Statement 1 is incorrect:** mRNA vaccines contain no virus, live or weakened. They deliver a piece of messenger RNA that instructs the body's cells to make a harmless viral protein (antigen), which trains the immune system.
- **Statement 2 is incorrect:** The mRNA does not enter the cell nucleus and cannot change a person's DNA; it is broken down by the cell after a short time.
- **Additional facts:** The mRNA is delivered inside tiny fat bubbles called lipid nanoparticles. The technology allows vaccines to be designed quickly and is being extended to personalised cancer vaccines.

Q 84. Ans. B

Explanation:

- Three-dimensional (3D) bioprinting is an advanced manufacturing technology that creates living tissues and biological structures by depositing cells, biomaterials, and growth factors layer by layer. The materials used, known as bio-inks, contain living cells embedded in supportive matrices.
- The technology has promising applications in tissue

engineering, regenerative medicine, drug testing, and the future development of transplantable organs such as skin, cartilage, and blood vessels.

- **Additional facts:** Bioprinting is used to make tissue for drug testing and research and aims one day to produce transplantable organs. It differs from organ-on-chip devices, which mimic organ function on a small chip.

Q 85. Ans. D

Explanation:

- **Statement 1 is incorrect:** A sodium-ion battery uses sodium ions, not lithium ions, which move between the electrodes to store and release charge.
- **Statement 2 is incorrect:** Sodium is far more abundant and cheaper than lithium, as it can be obtained from common salt and seawater; this is the main attraction of sodium-ion batteries.
- **Additional facts:** Sodium-ion batteries are seen as a lower-cost alternative to lithium-ion batteries for grid storage and some vehicles, though they currently store somewhat less energy for their weight.

Q 86. Ans. B

Explanation:

- Small Modular Reactors (SMRs) are advanced nuclear reactors that generate electricity through the process of nuclear fission. In this process, the nucleus of heavy atoms such as Uranium-235 splits into smaller nuclei, releasing a large amount of energy in the form of heat.
- This heat is used to produce steam, which drives turbines connected to electricity generators. SMRs are called “small” because they have a lower power output than conventional nuclear reactors and “modular” because many of their components can be factory-fabricated and assembled on-site.
- **Additional facts:** SMRs are promoted as cheaper, quicker to build, and easier to site, including for powering data centres. India is developing its own small reactor designs.

Q 87. Ans. C

Explanation:

- **Statement 1 is correct:** 6G, the planned successor to 5G, is expected to use much higher frequencies, reaching into the terahertz band, to achieve far greater data rates.
- **Statement 2 is correct:** Higher-frequency waves can carry more data but have a shorter range and

are blocked more easily by obstacles, so they need many small base stations placed close together.

- **Additional facts:** 6G is expected to support applications such as immersive holographic communication and very large sensor networks. Its standards are still under research, with deployment expected around 2030.

Q 88. Ans. B

Explanation:

- A digital twin is a virtual, real-time replica of a physical object, process, or system, kept up to date with data from sensors so that it mirrors the behaviour of the real one.
- Engineers use it to monitor performance, test changes, and predict faults without disturbing the actual system.
- The other options, namely a disk backup, a biological clone, and a fake account, are unrelated to this meaning.
- **Additional facts:** Digital twins are used for engines, factories, power grids, and even whole cities, often combined with the Internet of Things and artificial intelligence.

Q 89. Ans. C

Explanation:

- LiDAR (Light Detection and Ranging) measures distance by sending out pulses of laser light and timing how long the reflected pulses take to return.
- It is the optical counterpart of radar, which uses radio waves, and of sonar, which uses sound waves; magnetic fields are not used for ranging.
- **Additional facts:** LiDAR builds detailed three-dimensional maps and is widely used in self-driving cars, robotics, archaeology, and terrain mapping. It works in darkness, as it provides its own light.

Q 90. Ans. A

Explanation:

- Quantum dots are extremely small semiconductor nanocrystals, typically measuring 2–10 nanometres in diameter. Due to the phenomenon of quantum confinement, they emit different colours of light depending on their size smaller dots emit bluer light, while larger dots emit redder light.
- They are widely used in QLED displays, medical imaging, solar cells, biosensors, and quantum technologies because of their high brightness, colour purity, and tunable optical properties.

Q 91. Ans. A

Explanation:

- Gurjan, Irupu, Poon and Toon are characteristic tree species associated with Tropical Evergreen Forests of India. These forests occur in regions receiving heavy rainfall (generally over 200 cm annually), such as the Western Ghats, North-Eastern India, and the Andaman and Nicobar Islands.

Q 92. Ans. C

Explanation:

- **Goal 4 (Quality education):** This goal ensures inclusive and equitable quality education and promotes lifelong learning opportunities for all.
- **Goal 8 (Decent work and economic growth):** It aims to promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.
- **Goal 6 (1. Clean water and sanitation):** This focus area ensures the availability and sustainable management of water and sanitation for all.
- **Goal 14 (Life below water):** It targets the conservation and sustainable use of the oceans, seas, and marine resources for sustainable development.

Other Sustainable Development Goal:

Goal Focus Area Goal

- | | | |
|----|---|------|
| 1 | No Poverty | Goal |
| 2 | Zero Hunger | Goal |
| 3 | Good Health and Well-being | Goal |
| 5 | Gender Equality | Goal |
| 7 | Affordable and Clean Energy | Goal |
| 9 | Industry, Innovation and Infrastructure | Goal |
| 10 | Reduced Inequalities | Goal |
| 11 | Sustainable Cities and Communities | Goal |
| 12 | Responsible Consumption and Production | Goal |
| 13 | Climate Action | Goal |
| 15 | Life on Land | Goal |
| 16 | Peace, Justice and Strong Institutions | Goal |
| 17 | Partnerships for the Goals | |

Q 93. Ans. B

Explanation:

- **Statement 1 is correct:** Infrastructure projects (like highways, metro rails, airports, or waste management plants) typically suffer from long gestation periods and require massive upfront capital investment. VGF provides financial support traditionally as a capital grant during the construction stage to private sector entities to make these high-cost projects investment-friendly.

- **Statement 2 is incorrect:** The statement presents the conditions exactly backward. VGF is specifically meant for infrastructure projects that are economically justified but financially unviable.

- ✓ **Economically Justified:** The project provides immense social and economic benefits to the public (e.g., a rural highway or a clean water supply system).
- ✓ **Financially Unviable:** Due to user fee caps, high operating costs, or low initial traffic, the project cannot generate enough commercial revenue on its own to recover costs and attract private bidders.

Q 94. Ans. D

Explanation:

- The MCX iCOMDEX series consists of a composite index, two sectoral indices (Base Metals Index and Bullion Index), and four single commodity indices: Gold, Silver, Copper, and Crude Oil. marketscreener
- Natural Gas is notably absent from the iCOMDEX single commodity index series. While Natural Gas is traded on MCX, it is not included as a standalone index within the iCOMDEX series. The iComdex Energy Index covers crude oil and natural gas as a sectoral index, but Natural Gas does not have its own dedicated iCOMDEX index

Q 95. Ans. D

Explanation:

- **Statement 1 is correct:** Under the GST framework, a registered taxpayer can claim ITC on tax paid on almost all inward supplies of goods or services used in the course or furtherance of business. However, Section 17(5) of the CGST Act prescribes a specified negative list (often called blocked credits) where ITC cannot be claimed, even if used for business. Examples include motor vehicles (with exceptions), food and beverages, outdoor catering, beauty treatment, and goods lost, stolen, or destroyed.
- **Statement 2 is correct:** GST provides a specific mechanism called the Input Service Distributor (ISD). The ISD is an office of the supplier of goods or services which receives tax invoices issued towards input services. It then facilitates the distribution of this credit of CGST, SGST, UTGST, or IGST to its manufacturing units or group entities that actually consume those services, ensuring seamless flow of credit within corporate groups.

Q 96. Ans. C

Explanation:

- Anyonya Co-operative Bank Limited (ACBL) was established in 1889 in Vadodara (Baroda), Gujarat, and is widely regarded as India's first co-operative bank. It was founded by Vithal Laxman Kavthekar (also known as Bhausahab Kavthekar) with the primary aim of providing affordable credit to ordinary people and weaning them away from moneylenders.

Q 97. Ans. A

Explanation:

- **Statement 1 is correct:** One of the core guiding pillars of the policy is to create and exploit synergies among different forms of intellectual property, their corresponding statutes, and the various government agencies involved. It acts as a single, overarching vision document to move away from fragmented, siloed operations.
- **Statement 2 is correct:** The policy explicitly sets in place an institutional mechanism for implementation, monitoring, and review. The Department for Promotion of Industry and Internal Trade (DPIIT), under the Ministry of Commerce and Industry, is designated as the nodal department to coordinate and oversee future IPR developments. Additionally, a specialized professional body—the Cell for IPR Promotion and Management (CIPAM)—was created under DPIIT to act as the single point of reference for execution.
- **Statement 3 is incorrect:** The policy does not aim to replace existing IPR laws with a single, unified legal code. Instead, it legally recognizes that India already possesses a robust, well-established, and WTO TRIPS-compliant legislative and judicial framework. Its objective is to modify, strengthen, and update existing individual statutes (like the Patents Act, Trademarks Act, and Copyright Act) to remove procedural bottlenecks and balance property rights with larger public interest—not to dissolve them into a unified code.

Q 98. Ans. C

Explanation:

- FEMA (Foreign Exchange Management Act, 1999) replaced the earlier FERA (Foreign Exchange Regulation Act, 1973) and represented a fundamental philosophical shift in how foreign exchange violations are treated:

- Under FERA, foreign exchange violations were criminal offences, the burden of proof lay on the accused, and violations could lead to imprisonment.
- Under FEMA, violations are treated as civil offences, penalties are largely monetary/financial in nature, managed through adjudication by the Enforcement Directorate (ED) under a civil framework.

Q 99. Ans. D

Explanation:

- **Statement 1 is correct:** The demand function precisely shows the functional relationship between quantity demanded and the price of a good — expressed as $Q_d = f(P)$. This is the standard definition of the individual demand function, showing how quantity demanded varies with price.
- **Statement 2 is correct:** The demand function is derived under the ceteris paribus assumption — meaning all other determinants of demand are held constant. These include:
 - ✓ Income of the consumer
 - ✓ Tastes and preferences
 - ✓ Prices of related goods (substitutes and complements)
 - ✓ Expectations, population, etc.

Q 100. Ans. A

Explanation:

- **Statement 1 is correct:** Price elasticity of demand is precisely defined as the responsiveness (or sensitivity) of quantity demanded to a change in the price of that good. This is its standard textbook definition.
- **Statement 2 is correct:** The formula for price elasticity of demand is:
 - ✓ $E_d = \% \text{ change in quantity demanded} \div \% \text{ change in price}$
- This is the standard method of calculation, giving a coefficient that indicates whether demand is elastic, inelastic, or unitary.
- **Statement 3 incorrect:** Describes income elasticity of demand, not price elasticity of demand. Income elasticity measures the responsiveness of demand to a change in the consumer's income, calculated as:
 - ✓ $E_y = \% \text{ change in quantity demanded} \div \% \text{ change in income}$
- This is a completely separate concept. Conflating price elasticity with income elasticity is a classic exam trap.

Q 101. Ans. D

Explanation:

- The Marginal Product of Labour (MPL) measures the additional output produced by employing one additional unit of labour while keeping other factors, particularly capital, constant.
 - ✓ ΔQ = Change in output
 - ✓ ΔL = Change in labour
- Thus, MPL is the change in output divided by the change in labour.

Q 102. Ans. A

Explanation:

- **Statement 1 is correct:** Constant Returns to Scale means that when all inputs are increased by a certain proportion, output increases by the same proportion. For example, if both labour and capital are doubled, output exactly doubles. This is the standard definition of CRS.
- **Statement 2 is incorrect:** This is the critical trap. When only a single input is increased while others remain constant, that is the concept underlying the Law of Variable Proportions (Law of Diminishing Marginal Returns) not returns to scale. Returns to scale by definition always involves proportional changes in ALL inputs simultaneously.
- **Statement 3 is incorrect:** CRS is indeed associated with proportional change in all factors of production, this is essentially restating the definition from a different angle. All factors change in the same proportion, and output changes in that same proportion.

Q 103. Ans. D

Explanation:

- **Statement 1 is correct:** The services sector continues to be the primary engine of India's economic growth. According to the survey, it is estimated to grow by 9.1% in FY26, up from 7.2% in FY25. Driven by modern, tradable, and digitally delivered services, its share in nominal GDP increased to 53.6% in H1 FY26, while its share in Gross Value Added (GVA) reached a historic high of 56.4%.
- **Statement 2 is correct:** The Central Government has maintained a strong emphasis on capital-led growth over revenue expenditure. The survey notes that effective capital expenditure—comprising budgetary capital expenditure and grants-in-aid for the creation of capital assets—rose from a pre-pandemic average of 2.7% of GDP to 4.0% of GDP in FY25 (Provisional Actuals).

- **Statement 3 is correct:** India has made significant progress in fiscal consolidation in line with its post-pandemic roadmap. After reducing the fiscal deficit to 4.8% of GDP in FY25, the Union Budget targeted a further decline to 4.4% of GDP in FY26, thereby fulfilling the government's objective of bringing the fiscal deficit below 4.5% of GDP by FY26.

Q 104. Ans. D

Explanation:

- **Statement 1 is correct:** In line with the medium-term fiscal roadmap, the Fiscal Deficit in the Budget Estimates (BE) for 2026–27 is projected at 4.3% of GDP. This represents an improvement over the 4.4% of GDP estimated in the Revised Estimates (RE) for 2025–26 and is consistent with the government's objective of gradually stabilising the debt trajectory, with the Centre's debt-to-GDP ratio budgeted to decline to 55.6% in FY 2026–27.
- **Statement 2 is correct:** On the receipts side, the government has projected robust growth in tax collections. Out of the estimated non-debt receipts of ₹36.5 lakh crore in BE 2026–27, the Centre's net tax receipts are pegged at ₹28.7 lakh crore (approximately ₹28.67 lakh crore), reflecting a growth of about 7.2% over the Revised Estimates of 2025–26.

Q 105. Ans. B

Explanation:

- **Statement 1 is incorrect:** The scheme was not unveiled by the Ministry of Health and Family Welfare, nor is it part of the National Health Mission. It falls under the administrative ambit of the Ministry of Chemicals and Fertilizers (Department of Pharmaceuticals).
- **Statement 2 is correct:** Announced in the Union Budget 2026–27, the Government has officially backed this dedicated initiative with a robust financial allocation of ₹10,000 crore spread over a horizon of five years.
- **Statement 3 is correct:** To nurture long-gestation medical technology pipelines, the scheme explicitly targets early-stage innovation funding. It channels structured equity assistance and grant instruments (such as the Biopharma Discovery Grant Fund and Discovery & Development Equity Fund) to help startups and industrial entities push promising therapeutic candidates through critical clinical development milestones.

Q 106. Ans. D**Explanation:**

- **Vitamin B₁ (Thiamine) → Beriberi**
 - ✓ Thiamine is essential for carbohydrate metabolism and proper nerve function. Its deficiency causes Beriberi, characterized by muscle weakness, nerve damage, and cardiovascular complications.
- **Vitamin C (Ascorbic Acid) → Scurvy**
 - ✓ Vitamin C is required for collagen synthesis and wound healing. Deficiency leads to Scurvy, causing bleeding gums, poor wound healing, and joint pain.
- **Vitamin D → Rickets**
 - ✓ Vitamin D facilitates the absorption of calcium and phosphorus from the intestine. Deficiency in children results in Rickets, marked by soft and deformed bones.
- **Iron → Anaemia**
 - ✓ Iron is a key component of haemoglobin, which transports oxygen in the blood. Iron deficiency causes Anaemia, leading to fatigue, weakness, and reduced oxygen-carrying capacity.

Q 107. Ans. B**Explanation:**

- **Pair 1 is correctly matched: Forelimbs of a human and a whale → Homologous**
 - ✓ They share the same basic bone structure but perform different functions, indicating common ancestry.
- **Pair 2 is incorrectly matched: Wings of a bird and an insect → Homologous**
 - ✓ They are analogous, having different structures but the same function of flight.
- **Pair 3 is correctly matched: Appendix in humans → Vestigial organ**
 - ✓ It is a reduced, functionless remnant of an organ that was useful in ancestors.
- **Pair 4 is incorrectly matched: Wings of a bat and a bird → Analogous**
 - ✓ Both are modified forelimbs with the same basic structure, hence homologous.
- **Additional facts:** Homologous organs indicate divergent evolution from a common ancestor, while analogous organs indicate convergent evolution. Fossils provide direct evidence of evolution.

Q 108. Ans. C**Explanation:**

- The sex of a child is determined by the sperm of the

father. Of the 23 pairs of human chromosomes, one pair is the sex chromosomes.

- A woman is XX, so every egg carries an X chromosome; a man is XY, so half his sperm carry an X and half carry a Y.
- If an X-bearing sperm fertilises the egg, the child is a girl (XX); if a Y-bearing sperm does so, the child is a boy (XY). Thus the father, not the mother, determines the sex.
- **Additional facts:** It is therefore scientifically wrong to blame the mother for the sex of a child. The Y chromosome carries the male-determining genes.

Q 109. Ans. C**Explanation:**

- **Statement 1 is correct:** Lactobacillus bacteria convert the lactose in milk into lactic acid, turning milk into curd.
- **Statement 2 is correct:** Yeast (Saccharomyces) is used in the large-scale production of alcohol through the fermentation of sugars.
- **Statement 3 is incorrect:** Penicillin is obtained from a fungus, the mould Penicillium, discovered by Alexander Fleming, not from a bacterium.
- **Statement 4 is correct:** Rhizobium bacteria in the root nodules of leguminous plants fix atmospheric nitrogen, enriching the soil.
- **Additional facts:** Microorganisms can also be harmful, causing diseases such as tuberculosis (bacteria), malaria (a protozoan), and influenza (a virus). Pasteurisation kills harmful microbes in milk.

Q 110. Ans. D**Explanation:**

- Geotropism (gravitropism) is the directional growth of a plant part in response to gravity; roots show positive geotropism (growing downward) and shoots show negative geotropism (growing upward).
- Phototropism is growth in response to light, hydrotropism is growth towards water, and thigmotropism is growth in response to touch, as in the tendrils of climbers.
- **Additional facts:** These directional growth movements are controlled chiefly by the hormone auxin and are slow, unlike the rapid movements seen in the touch-me-not plant (Mimosa).

Q 111. Ans. B**Explanation:**

- **Statement 1 is incorrect:** Cabinet committees are extra constitutional in emergence, as the

Constitution does not mention them; they are established under the Rules of Business framed by the government for the transaction of its work.

- **Statement 2 is correct:** Cabinet committees are set up by the Prime Minister according to the requirements of the time and the exigencies of the situation, and their number, nomenclature, and composition vary from time to time.
- **Additional Information:** They are of two types, standing and ad hoc, the former being permanent and the latter temporary. Their membership usually varies from three to eight, they normally include only cabinet ministers but non cabinet ministers may also be members, and they are presided over by the Prime Minister or another senior minister. Important standing committees include the Cabinet Committee on Political Affairs, the Cabinet Committee on Economic Affairs, the Cabinet Committee on Security, and the Appointments Committee of the Cabinet, and they sort out issues and formulate proposals for the consideration of the cabinet, sometimes taking decisions themselves which the cabinet may review.

Q 112. Ans. B

Explanation:

- **Statement 1 is incorrect:** The State Chief Information Commissioner and the State Information Commissioners are appointed by the Governor, not the President, on the recommendation of a committee consisting of the Chief Minister as chairperson, the leader of the opposition in the legislative assembly, and a state cabinet minister nominated by the Chief Minister.
- **Statement 2 is correct:** After the Right to Information Amendment Act of 2019, the term of office, salary, allowances, and other conditions of service of the State Chief Information Commissioner and the State Information Commissioners are prescribed by the Central Government, whereas earlier the Act itself fixed a term of five years.
- **Additional Information:** The State Information Commission is a statutory body created under the Right to Information Act of 2005 and consists of the State Chief Information Commissioner and not more than ten State Information Commissioners. It hears complaints and decides appeals concerning state public authorities, and in the exercise of its functions it has the powers of a civil court, including summoning witnesses and requiring the production of documents.

Q 113. Ans. C

Explanation:

- **Statement 1 is correct:** Political parties are registered with the Election Commission under Section 29A of the Representation of the People Act of 1951, and registration entitles a party to certain benefits even before it secures recognition as a state or national party.
- **Statement 2 is correct:** The reservation and allotment of election symbols are governed by the Election Symbols Reservation and Allotment Order of 1968, under which the Election Commission allots reserved symbols to recognised parties and free symbols to others.
- On the basis of poll performance, the Commission recognises parties as national or state parties, and recognised parties enjoy benefits such as a reserved symbol, free broadcast time on state media, access to electoral rolls, and a greater number of star campaigners. Party systems are generally classified as one party, two party, or multi party systems, and India has a multi party system marked by a large number of regional and national parties. The Commission has the exclusive power to settle disputes over symbols when a recognised party splits.

Q 114. Ans. C

Explanation:

- **Statement 1 is correct:** The National Commission to Review the Working of the Constitution was set up in the year 2000 by a resolution of the Government of India to examine, in the light of the experience of fifty years, how far the existing Constitution could respond to the needs of an efficient, smooth, and effective system of governance.
- **Statement 2 is correct:** The Commission was chaired by M. N. Venkatachaliah, a former Chief Justice of India, and it was specifically directed to undertake its review within the framework of parliamentary democracy and without interfering with the basic structure of the Constitution.
- The Commission submitted its report in 2002 and made a large number of recommendations covering fundamental rights, directive principles, electoral and parliamentary reforms, the pace of socio economic change, and the strengthening of institutions, while concluding that the Constitution had stood the test of time and did not require a radical overhaul. Earlier efforts to review the working of the Constitution included the Swaran Singh Committee of 1976, whose recommendations led to the 42nd Amendment.

Q 115. Ans. C

Explanation:

- **Statement 1 is correct:** 'Mithila Makhana' received the Geographical Indication (GI) Tag in 2022. The GI tag recognizes the unique quality and traditional cultivation practices of Makhana grown in the Mithila region of Bihar.
- **Statement 2 is incorrect:** At a public rally in Samastipur, the Prime Minister described the newly inaugurated **National Makhana Board** as a "revolutionary step" for Makhana farmers. India contributes approximately **80% of the global demand for Makhana (Fox Nuts)**.
- **Statement 3 is correct:** Makhana, also known as **Fox Nut** or **Gorgon Nut**, is the dried edible seed of the prickly water lily **Euryale ferox**. It is cultivated primarily in **tropical and subtropical climatic conditions**. Makhana is popularly known as the "**Black Diamond**" because of its small, round black or brown seeds, which turn white after processing. **Bihar accounts for nearly 90% of India's total Makhana production**. The major Makhana-producing districts include **Darbhanga, Madhubani, Purnea, Katihar, Saharsa, Supaul, Araria, Kishanganj, and Sitamarhi**.

Q 116. Ans. C

Explanation:

- The **Kisan Credit Card (KCC) Scheme** was developed by **NABARD (National Bank for Agriculture and Rural Development)** to provide timely and adequate institutional credit to farmers for meeting their agricultural and allied sector requirements. The scheme was officially launched in **August 1998** by public sector banks based on the recommendations of the **R. V. Gupta Committee** on agricultural credit. Over the years, the scheme has been expanded to improve farmers' access to affordable credit. In **December 2020**, Prime Minister **Narendra Modi** launched a special drive to extend KCC coverage, particularly to beneficiaries of the **PM-KISAN** scheme. Further strengthening the scheme, the Government increased the **collateral-free loan limit under KCC from ₹3 lakh to ₹5 lakh in April 2025**, enabling farmers to avail higher credit at concessional interest rates for agricultural and allied activities.

Q 117. Ans. D

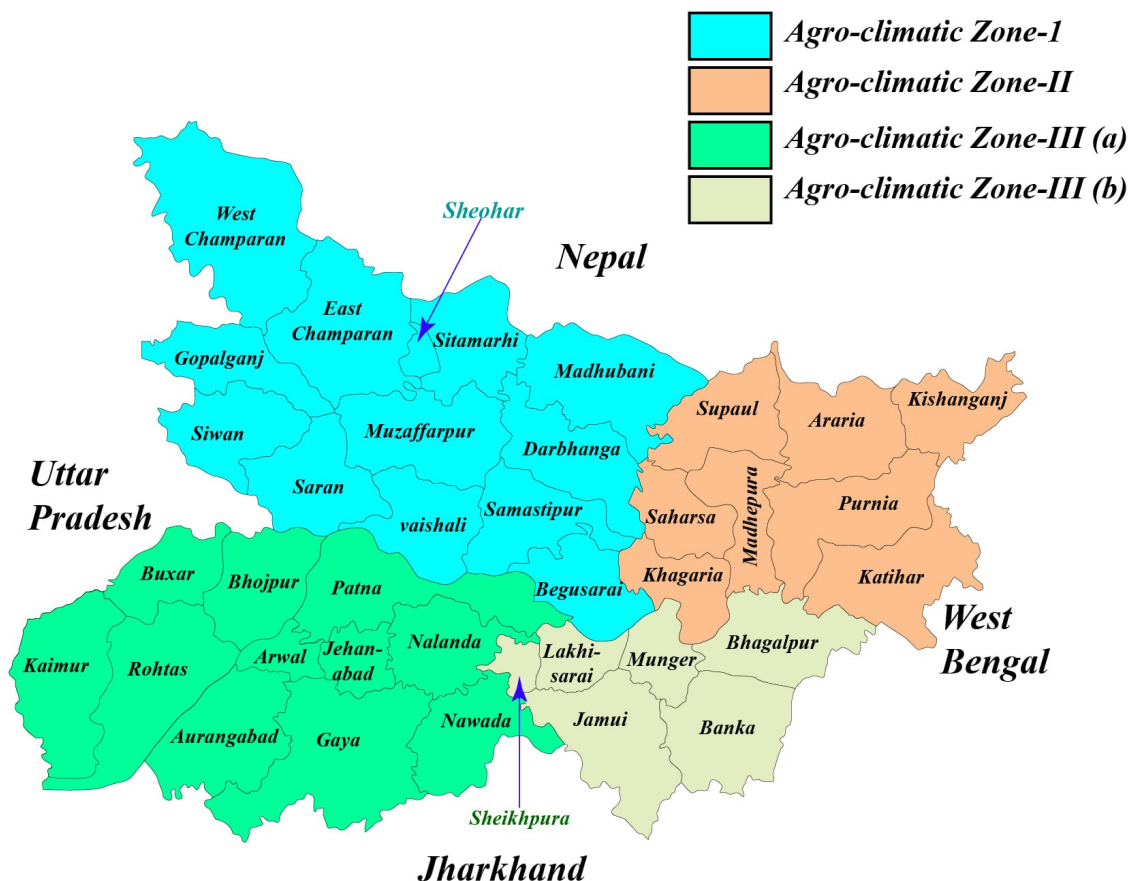
Explanation:

- **Bihar Veterinary College, Patna** was established on **2 April 1927** by **Sir Henry Wheeler**, the then Governor of Bihar and Orissa, and commenced its academic activities on **7 April 1927**. The establishment of the college is associated with the recommendations of the **Ganesh Dutt Committee**. Subsequently, with the enforcement of the **Bihar Agricultural University Act, 1971**, Bihar Veterinary College became one of the pioneering constituent colleges of **Rajendra Agricultural University, Pusa**. Following its integration into the agricultural university system, all fourteen departments of the college began functioning under the university framework, contributing significantly to veterinary education, research, and extension services in the state. Today, the college functions under the **Bihar Animal Sciences University (BASU), Patna**, and remains one of the premier veterinary institutions in India.

Q 118. Ans. D

Explanation:

- **Assertion (A) is false** In Bihar, agricultural or agro-climatic zoning is **not based on climate and soil independently**. Rather, it is based on the **combined influence** of climatic factors (rainfall, temperature, humidity, length of growing period) and edaphic factors (soil type, fertility, drainage, texture). These factors interact to determine crop suitability, productivity, and farming systems.
- **Reason (R) is true:** Agro-climatic zones are defined using a combination of rainfall, temperature, terrain, and soil characteristics. Agro-climatic classification in Bihar, as adopted by agricultural planning agencies, considers multiple parameters, including: Rainfall distribution, Temperature regime, Physiography/terrain, Soil characteristics, Water availability.
- **Why is the answer (D)**
 - ✓ Assertion (A) is **false** because climate and soil are **interdependent and jointly considered** in agricultural zoning.
 - ✓ Reason (R) is **true** because agro-climatic zones are indeed delineated using a combination of climatic and soil-related parameters.



Agro-climatic Zones of Bihar

Q 119. Ans. C

Explanation:

- The Ganga River is the most important river of Bihar and divides the state into two distinct geographical regions—North Bihar and South Bihar. Flowing from west to east for about 445 km within the state, it serves as the principal drainage artery of Bihar. The river enters Bihar near Chausa in Buxar district and flows eastward through the fertile alluvial plains before entering Jharkhand near Manihari in Katihar district. Along its course, it passes major urban centres such as Patna, Munger, Bhagalpur, and Kahalgaon. The Ganga basin in Bihar is enriched by numerous tributaries, including seven major Himalayan rivers from the north and six peninsular rivers from the south, making the region highly fertile but also susceptible to recurrent floods.
- The 12 districts situated along the banks of the Ganga in Bihar are Buxar, Bhojpur, Saran, Patna, Vaishali, Samastipur, Begusarai, Lakhisarai, Munger, Khagaria, Bhagalpur, and Katihar. Among these, Patna district has the longest stretch of the Ganga, approximately 99 km. Several districts such as Bhojpur, Vaishali, Samastipur, Begusarai, and Lakhisarai are directly influenced by the river's floodplain processes, while districts like Nalanda, Banka, and Purnia are not located on the banks of the Ganga, although they are influenced by other river systems and basin characteristics.



Q 120. Ans. D

Explanation:

- Pair 1 is correct: Jayanti and Tumuni are recognized tributaries of the Ajay River. The Ajay originates in Jamui (Bihar) and flows through Jharkhand and West Bengal. Jayanti joins it in Jharkhand, while Tumuni joins it in West Bengal.
- Pair 2 is correct: Both Chandraprabha and Durgavati are tributaries of the Karmanasa River. Durgavati is an important right-bank tributary flowing through Kaimur district of Bihar before joining the Karmanasa.
- Pair 3 is correct: Morhar (Mohar) is a well-recognized tributary of the Punpun. However, Dordha is generally treated as a tributary/sub-tributary within the Punpun basin in regional records. Major tributaries commonly cited are Morhar (Mohar), Madar, and Butane.
- Pair 4 is correct: Fariani and Dhemana Dhar are minor tributary channels associated with the Kosi river system in Bihar's Kosi basin. They are not among the major Himalayan tributaries but occur in Bihar river-system records.

Q 121. Ans. D

Explanation:

- Kanwar Lake (Kabartal Lake), located in the Manjhaul area of Begusarai district, Bihar, is a large freshwater residual oxbow lake formed by the meandering and shifting course of the Burhi Gandak River, a tributary of the Ganga. The lake extends across the floodplains around Manjhaul town and the nearby village of Kanwar. Owing to its rich biodiversity and wetland ecosystem, it serves as an important habitat for numerous resident and migratory bird species and functions as a significant stopover site along the Central Asian Flyway.

Q 122. Ans. D

Explanation:

- Assertion (A) is False The North-East Alluvial Plain is actually more vulnerable to floods than the South-West Alluvial Plain. This region includes districts such as Purnea, Katihar, Araria, Kishanganj, Saharsa, Supaul, and Madhepura, which are traversed by highly flood-prone rivers like the Kosi, Mahananda, and their tributaries.

- **Reason (R) is true:** The North-East Alluvial Plain receives the **highest annual rainfall in Bihar**, generally ranging between **1200 mm and 1700 mm or more**, particularly in districts close to the Himalayan foothills.
- Why is the answer (D)?
 - ✓ Assertion (A) is **incorrect** because the North-East Alluvial Plain is one of the **most flood-prone regions of Bihar**.
 - ✓ Reason (R) is **correct** because this zone indeed receives the highest rainfall among Bihar's agro-climatic zones.

Q 123. Ans. B

Explanation:

- According to the Bihar Economic Survey 2025–26, the per capita energy consumption in Bihar increased to 374 kilowatt-hours (kWh) in 2024–25. The survey highlights that Bihar's per capita energy consumption has risen steadily from 134 kWh in 2011–12 to 374 kWh in 2024–25, indicating significant growth in electricity access and demand over the last decade. AT&C (Aggregate Technical & Commercial) losses declined from 42.99% in 2013–14 to 15.54% in 2024–25. Bihar's DISCOMs (North Bihar Power Distribution Company Limited and South Bihar Power Distribution Company Limited) received Innovation & Impact Awards at the 19th India Energy Summit 2025.

Q 124. Ans. B

Explanation:

- **Statement 1 is correct:** Census is a Union subject under Entry 69 of List I (Union List) of the Seventh Schedule to the Constitution of India. It is conducted by the **Office of the Registrar General and Census Commissioner, India**, functioning under the **Ministry of Home Affairs**. While the Union Government is responsible for planning, supervision, and coordination of the Census, the enumeration work is carried out through a vast administrative machinery involving officials and staff of **State Governments and Union Territory Administrations** appointed under the provisions of the Census Act, 1948.
- **Statement 2 is incorrect:** Census 2027 will be conducted in two phases. Phase I, the **Houselisting and Housing Census (HLO)**, will be carried out between **April and September 2026** for a period of **30 days in each State/UT**, with schedules finalized in consultation with the respective governments.

This phase will collect data on **housing conditions, household amenities, and assets**. Phase II, the **Population Enumeration (PE)**, is scheduled for **February 2027** and will gather detailed information on individuals, including their **demographic, socio-economic, educational, cultural, migration, and fertility-related characteristics**.

S No.	Question	S No.	Question
1	Line Number	9	Household number
2	Building number	10	Total number of persons normally residing in the household
3	Census house number	11	Name of the head of the household
4	Predominant material of Floor of the Census house	12	Sex of the head of the household
5	Predominant material of Wall of the Census house	13	Weather the head of the household belongs to SC/ST/Other
6	Predominant material of Roof of the Census house	14	Ownership status of the Census house
7	Ascertain use of Census house	15	Number of dwelling rooms exclusively in possession of the household
8	Condition of the Census House	16	Number of married couple(s) living in the household
17	Main source of drinking water	26	Radio/Transistor
18	Availability of drinking water source	27	Television
19	Main source of lighting	28	Access to Internet
20	Access to Latrine	29	Laptop/Computer
21	Type of Latrine	30	Telephone/Mobile Phone/Smartphone
22	Waste water outlet	31	Bicycle and Scooter/Motorcycle/Moped
23	Availability of bathing facility	32	Car/Jeep/Van
24	Availability of kitchen and LPG/PNG connection	33	Main cereal consumed in the household
25	Main Fuel used for cooking	34	Mobile number (For Census related communications only)

Q 125. Ans. A

Explanation:

- **Statement 1 is correct:** Bihar Rajya Beej Nigam Limited (BRBN) has introduced an innovative seed distribution system under which certified seeds are delivered directly to farmers at their homes across all districts of Bihar. Through its online application system, farmers can choose either to collect seeds from designated sale points or receive them through the **home delivery scheme**. The entire process, from application to distribution and feedback collection, is carried out digitally.
- **Statement 2 is incorrect:** Bihar is widely recognized as the **first state in India to implement large-scale home delivery of seeds to farmers**. The initiative was initially tested as a pilot project and was subsequently expanded across the state. It helps farmers obtain seeds directly at their doorsteps, saving both time and transportation costs.

Q 126. Ans. C

Explanation:

- The **Digha–Kankarbagh Integrated Wastewater Project** in Patna, implemented by the **Bihar Urban Infrastructure Development Corporation**

(BUIDCO), received the “Wastewater Project of the Year 2026” award at the Global Water Awards 2026. The award was presented on 19 May 2026 during the Global Water Summit 2026 held in Madrid, Spain. The Global Water Awards are organized annually by Global Water Intelligence (GWI), an internationally recognized water-sector intelligence and research organization, to honor excellence and innovation in the global water industry.

Q 127. Ans. A

Explanation:

- **Assertion (A)** is true South-West Alluvial Plain (Zone-IIIB) is more drought-prone than North-East Alluvial Plain (Zone-II). The **South-West Alluvial Plain (Zone-IIIB)**, comprising districts such as Buxar, Bhojpur, Rohtas, Kaimur, Aurangabad, Arwal, Patna, Nalanda, Jehanabad, and Gaya, is one of the relatively drier agro-climatic regions of Bihar. This zone frequently experiences **moisture stress, delayed monsoon effects, and drought-like conditions**, especially in years of deficient rainfall.
- **Reason (R)** is true: **Zone-IIIB** receives significantly lower rainfall and experiences more frequent water-deficit years. Rainfall in Bihar generally decreases from **north-east to south-west**. Zone-II (North-East Alluvial Plain) receives about **1200–1700 mm** of annual rainfall, while Zone-IIIB generally receives around **900–1200 mm**. Consequently, Zone-IIIB faces more frequent **water-deficit years**, lower soil moisture availability, and greater dependence on irrigation.
- Why is (R) the correct explanation of (A)?
 - ✓ Drought proneness is primarily determined by the **amount and reliability of rainfall**. Since Zone-IIIB receives substantially less rainfall and experiences greater moisture deficiency than Zone-II, it is naturally more vulnerable to drought conditions.

Q 128. Ans. A

Explanation:

- **Assertion (A)** is true: The Graded Response Action Plan (GRAP), implemented in Delhi-NCR under the directions of the Commission for Air Quality Management (CAQM), is indeed divided into four progressively stricter stages:
 - ✓ Stage I — Poor (AQI 201–300)
 - ✓ Stage II — Very Poor (AQI 301–400)
 - ✓ Stage III — Severe (AQI 401–450)
 - ✓ Stage IV — Severe+ (AQI above 450)

- **Reason (R)** is true: The operational trigger for activating any given stage of GRAP depends entirely on the prevailing ambient air quality. Each stage is strictly linked to a specific quantitative band of the Air Quality Index (AQI) recorded in the region.
- **Linkage ((R) explains (A))**: The reason why the plan is “graded” into four escalating tiers of interventions is precisely because air pollution does not spike uniformly. By linking the actions directly to worsening AQI thresholds, the Commission for Air Quality Management (CAQM) can implement measured, proportional emergency steps rather than enforcing blanket lockdowns prematurely.

Q 129. Ans. D

Explanation:

- The Union Cabinet approved the landmark Sabka Bima Sabki Raksha (Amendment of Insurance Laws) framework, completely liberalizing foreign equity participation by raising the Foreign Direct Investment (FDI) limit from 74% to 100% under the automatic route.
- India’s insurance market has undergone a highly controlled, phased liberalization over the past two decades. The FDI cap was initially pegged at 26% in 2001, increased to 49% in 2015, expanded to 74% in 2021, and has now reached 100% to permit full foreign ownership.

Q 130. Ans. A

Explanation:

- **Milk Production:**
 - ✓ Total milk production in the country is 247.87 million tonnes during 2024-25.
 - ✓ India ranks 1st in the world in terms of total milk production.
 - ✓ The milk production has increased by 3.58% over the previous year (2023-24).
 - ✓ The per-capita availability of milk is 485 grams per day
 - ✓ Uttar Pradesh is India’s leading milk-producing state followed by Rajasthan, Madhya Pradesh
- **Egg Production:**
 - ✓ The total egg production in the country is 149.11 billion numbers
 - ✓ India ranks 2nd in the world in terms of total Egg production
 - ✓ The egg production has increased by 4.44% as compared to previous year (2023-24).
 - ✓ Out of the total eggs produced, commercial poultry contributed 84.49%.

- ✓ Andhra Pradesh is the leading egg-producing state in India

● Meat Production:

- ✓ The total meat production in the country is 10.50 million tonnes during 2024-25.
- ✓ India ranks 4th in the world in terms of total Meat production.
- ✓ The meat production has increased by 2.46% as compared to previous year (2023-24).
- ✓ West Bengal is the leading producer of meat in India followed by Uttar Pradesh, Maharashtra.

● Wool Production:

- ✓ The total wool production in country for the year 2024-25 stands at 34.57 Million Kilograms.
- ✓ This represents a notable increase of 2.63% when compared to the previous year's figures, indicating a positive trend in the sector.
- ✓ India ranks 9th in the world in terms of total wool production
- ✓ Rajasthan is the leading producer of wool in India followed by Jammu & Kashmir

Fadnavis laid the foundation for Lloyds Metals & Energy Limited's first integrated steel plant in Vidarbha's Gadchiroli district a 4.5 million tonne per annum plant at Konsari, marking the first such facility in the mineral-rich but historically under-industrialised region.

● Gujarat → First Indian State to launch genome sequencing initiative

- ✓ Gujarat became the first state in India to launch a large-scale genome sequencing project, specifically focused on tribal communities, led by the Gujarat Biotechnology Research Centre (GBRC), aimed at building a reference genetic database to improve healthcare for tribal populations.

● Goa → First Ro-Ro service on Mandovi River

- ✓ Goa launched India's first RoRo (Roll-on/Roll-off) ferry service on the Mandovi River with ferries named Gangotri and Dwarka connecting Chorao Island with Ribandar, making travel faster for passengers and vehicles

Q 131. Ans. C

Explanation:

- Project Suncatcher is a research moonshot launched by Google Research designed to move data-intensive artificial intelligence (AI) and machine learning (ML) workloads off-planet and into low Earth orbit (LEO).
- The project details a modular network of compact satellites equipped with Google's custom-built AI chips, specifically the Trillium TPU v6e (Tensor Processing Unit). By operating in custom "dawn-dusk" sun-synchronous orbits, these orbital data centers can harvest near-constant sunlight, making solar capture up to eight times more efficient than Earth-based panels.

Q 132. Ans. D

Explanation:

- Patna → 6th Indian Open Athletics Meet
 - ✓ Patna hosted a major national-level sports competition for the first time on July 19, 2025, with the 6th Indian Open Athletics Meet held at the Patliputra Sports Complex in Kankarbagh. The Athletics Federation of India chose Bihar as the host state for this prestigious competition for the first time.
- Vidarbha → First Integrated Steel Plant
 - ✓ Maharashtra Chief Minister Devendra

Q 133. Ans. C

Explanation:

- Statement 1 is correct: The tournament was held in Riyadh, Saudi Arabia. It featured 16 world-class players representing 13 professional esports organizations competing in a rapid time control (10+0) format.
- Statement 2 is correct: Magnus Carlsen won the inaugural Chess Esports World Cup title by defeating Alireza Firouzja in the Grand Final. However, the statement says Firouzja belonged to Team Falcons — this needs verification. Carlsen represented Team Liquid and beat Alireza Firouzja of Team Falcons.

Q 134. Ans. B

Explanation:

- Row 1 is correctly matched: The prestigious Vigyan Ratna award for lifetime achievement was conferred posthumously on the legendary astrophysicist Prof. Jayant Vishnu Narlikar for his monumental contributions to Physics (specifically astrophysics and the Hoyle–Narlikar theory of gravity).
- Row 2 is correctly matched: Dr. Yusuf Mohammad Seikh was conferred the Vigyan Shri award for his impactful work and leadership in the domain of Atomic Energy.
- Row 3 is incorrectly matched: While Dr. Satendra Kumar Mangrauthia (Principal Scientist at

ICAR-IIRR) was indeed awarded the Vigyan Yuva-Shanti Swarup Bhatnagar award, his field of research is Agricultural Science (specifically rice biotechnology and climate-resilient agriculture), not Physics.

Q 135. Ans. C

Explanation:

- Punjab recently became the first State in India to successfully implement the amended BharatNet Scheme across its entire territory. The achievement marks a significant milestone in expanding rural digital infrastructure and strengthening broadband connectivity.
- BharatNet is the Government of India's flagship programme aimed at providing high-speed broadband connectivity to all Gram Panchayats, thereby bridging the digital divide and enabling access to e-governance, education, healthcare, financial services, and other digital services in rural areas.

Q 136. Ans. A

Explanation:

- The Inland Waterways Authority of India (IWAI) signed a series of MoUs worth ₹3,000 crore during India Maritime Week 2025 to strengthen inland waterways, industrial logistics, cargo movement, and passenger connectivity across the Northeastern region. The agreements aim to enhance sustainable transport and improve multimodal connectivity in states such as Assam, Mizoram, Arunachal Pradesh, Manipur, Tripura, Nagaland, and Meghalaya.

Q 137. Ans. D

Explanation:

- India's first MWh-scale Vanadium Redox Flow Battery (VRFB) system a 3 MWh capacity installation was inaugurated at NTPC NETRA (New and Renewable Energy Technology Research Authority), the R&D centre of NTPC located in Greater Noida. The system is significant for reducing India's overdependence on lithium and rare-earth imports for battery technology. It supports India's Long-Duration Energy Storage (LDES) goals for integrating renewable energy into the grid.

Q 138. Ans. B

Explanation:

- The Major Non-NATO Ally (MNNA) status is a designation granted by the United States to certain strategic partners.

- **Major Non-NATO Ally status:** Argentina, Australia, Bahrain, Brazil, Colombia, Egypt, Israel, Japan, Jordan, Kenya, Kuwait, Morocco, New Zealand, Pakistan, the Philippines, Qatar, Saudi Arabia, South Korea, Thailand, and Tunisia.
- UAE does not designated as MNNA. Despite being a close U.S. partner, UAE does not appear on the official MNNA list.

Q 139. Ans. B

Explanation:

- The Preah Vihear Temple, recently in the news due to cross-border military skirmishes between Cambodia and Thailand, is located in Cambodia, near the Cambodia-Thailand border atop the Dângrêk Mountains.
- Dedicated to Lord Shiva, the temple dates back to the 11th–12th centuries during the Khmer Empire and is renowned for its remarkable Khmer architecture. In 2008, it was inscribed as a UNESCO World Heritage Site. The area surrounding the temple has long been a source of territorial dispute between Cambodia and Thailand, occasionally leading to heightened tensions and military confrontations.

Q 140. Ans. B

Explanation:

- In the IUCN World Heritage Outlook 2025, Khangchendzonga National Park in Sikkim was the only Indian natural World Heritage Site to receive a "Good" rating, reflecting its strong conservation status, effective management, and ecological resilience. It is also notable for being India's first mixed UNESCO World Heritage Site, recognised for both its natural and cultural significance.

Q 141. Ans. A

Explanation:

According to the question,
The new number must be divisible by 12, 15, and 18.
Therefore,
 $\text{LCM}(12, 15, 18)$
 $= 2^2 \times 3^2 \times 5$
 $= 180$
Now,
 $15425 \div 180$
 $= 85$ with a remainder of 125
So, the next multiple of 180
 $= 180 \times 86$
 $= 15480$
The required minimum number

$$= 15480 - 15425$$

$$= 55$$

Thus, the correct answer is 55.

Q 142. Ans. C

Explanation:

According to the question,

To find the units digit, we focus only on the units digits.

$$81 \times 82 \times 83 \times 84 \times 85 \times 86 \times 87 \times 88 \times 89$$

$$\Rightarrow 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9$$

Both 2 and 5 are present in this product.

Therefore, the product will have a factor of 10, so the units digit will be 0.

Q 143. Ans. C

Explanation:

$$\text{Distance} = 648 \text{ km} = 648,000 \text{ m}$$

$$\text{Speed} = 18 \text{ m/sec}$$

$$\text{Time} = \text{Distance} / \text{Speed}$$

$$= 648,000 / 18$$

$$= 36,000 \text{ seconds}$$

Now,

$$36,000 \text{ seconds} = 36,000 / 3,600 = 10 \text{ hours}$$

Q 144. Ans. D

Explanation:

Ratio of time taken by Seema and Reema = 4 : 6

Ratio of work efficiency of Seema and Reema = 6 : 4

$$= 3 : 2$$

Therefore, the amount will also be distributed in the ratio of 3 : 2.

$$\text{Seema's share} = 3/5 \times 3600 = \text{Rs. } 2160$$

Q 145. Ans. B

Explanation:

Number of failed candidates = 35% of 2400

$$= 2400 \times 7/20$$

$$= 120 \times 7$$

$$= 840$$

$$\text{Number of passed candidates} = 2400 - 840 = 1560$$

Q 146. Ans. B

Explanation:

According to the question,

Selling price at 20% loss = Rs. 1200

$$\text{Hence, Purchase Price (CP)} = 1200 \times 100/80 =$$

$$\text{Rs } 1500$$

Now at 25% profit,

$$\text{Selling Price (SP)} = 1500 \times 125/100 = \text{Rs } 1875$$

Q 147. Ans. A

Explanation:

According to the question,

The two numbers are $4x$ and $7x$.

Their difference = 45

$$\Rightarrow 7x - 4x = 45$$

$$\Rightarrow 3x = 45$$

$$\Rightarrow x = 15$$

Therefore, the numbers are:

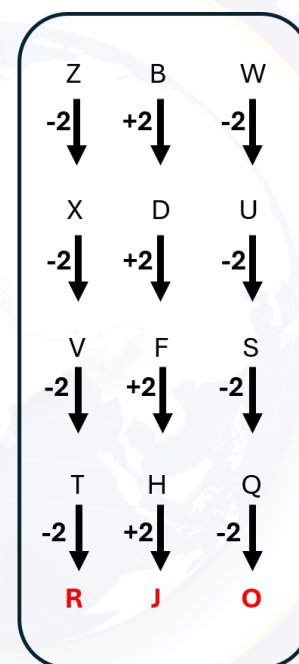
$$\text{First} = 4 \times 15 = 60$$

$$\text{Second} = 7 \times 15 = 105$$

$$\text{Sum of the numbers} = 60 + 105 = 165$$

Q 148. Ans. B

Explanation:



Q 149. Ans. D

Explanation:

Jupiter, Saturn, and Mars are planets, whereas the Moon is a natural satellite.

Q 150. Ans. A

Explanation:

Given:

$M @ N \rightarrow M$ is the father of N

$M \# N \rightarrow M$ is the mother of N

$M \$ N \rightarrow M$ is the brother of N

$M \% N \rightarrow M$ is the sister of N

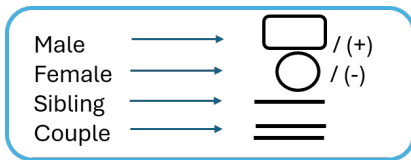
A: $A \$ B \# D$

$A \$ B \Rightarrow A$ is the brother of B

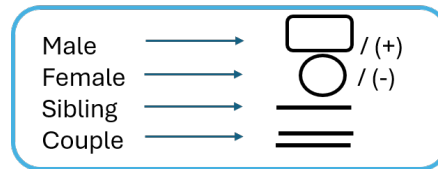
$B \# D \Rightarrow B$ is the mother of D

Since B is the mother of D and A is B 's brother.

Therefore, A is the maternal uncle of D .



A(+) — B(-)
|
D



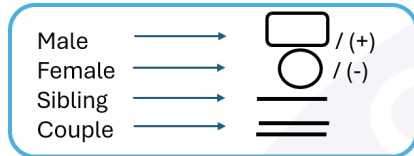
A(+)
|
B(-)
|
D

B. A % B # D

A is the sister of B.

B is the mother of D.

A becomes maternal aunt, not uncle.



A(-) — B(-)
|
D

D. A \$ B @ D

A is the brother of B.

B is the father of D.

A becomes paternal uncle, not maternal uncle.



A(+) — B(+)
|
D

C. A @ B # D

A is the father of B.

B is the mother of D.

A becomes maternal grandfather of D.

Hence, Option A is the correct answer.

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